

Crisis and Commodities: Analyzing the Correlation between Commodity Prices and Financial Markets during Major Global Shocks

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Lausanne, Switzerland

January 2025

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Abstract

Financial markets and commodity prices have historically exhibited complex interdependencies that become particularly pronounced during periods of global crisis. Major shocks, such as the 2008 Global Financial Crisis, the COVID-19 pandemic, and the Russia-Ukraine conflict, have demonstrated how commodities can serve simultaneously as safe-haven assets, inflation hedges, and sources of systemic risk. Understanding these dynamics is essential for investors, policymakers, and corporations seeking to manage portfolio risk, hedge against volatility, and anticipate market behavior during periods of stress. This thesis aims to provide a comprehensive empirical analysis of how commodity markets correlate with broader financial markets during major disruptions, with the goal of identifying patterns that can inform more resilient investment strategies.

Les marchés financiers et les prix des matières premières ont historiquement présenté des interdépendances complexes, qui deviennent particulièrement marquées en période de crise mondiale. Des chocs majeurs, tels que la crise financière mondiale de 2008, la pandémie de COVID-19 et le conflit entre la Russie et l'Ukraine, ont montré que les matières premières peuvent simultanément agir comme des valeurs refuges, des instruments de couverture contre l'inflation et des sources de risque systémique. Comprendre ces dynamiques est essentiel pour les investisseurs, les décideurs publics et les entreprises cherchant à gérer le risque de portefeuille, à se protéger contre la volatilité et à anticiper le comportement des marchés en période de tensions. Cette thèse vise à fournir une analyse empirique approfondie des corrélations entre les marchés des matières premières et les marchés financiers au sens large lors de perturbations majeures, afin d'identifier des schémas susceptibles d'éclairer des stratégies d'investissement plus résilientes.

Keywords: Commodity Markets, Financial Markets, Crisis, Volatility, Risk

JEL Classification: G11 - G13 - G17 - Q31

Acknowledgements

I would like to express my sincere gratitude to all those who have supported me throughout the completion of this master's thesis. First and foremost, I am deeply grateful to my university supervisor, Giuseppe Balocchi, for his invaluable guidance, expertise, and continuous encouragement throughout this research journey. He was always available to provide support, answer my questions, and offer insightful feedback, which proved essential in shaping both the direction and the quality of this work.

I am equally grateful to Crédit Agricole (Indosuez Wealth Management) and, in particular, to the Global Commodity Finance (GCF) team for their support throughout my internship. Working within such a professional and collaborative environment was a highly rewarding experience. I sincerely thank the team for their guidance, availability, and trust, and for making this internship both enriching and enjoyable.

My heartfelt thanks go to my family and friends for their unwavering support, patience, and understanding throughout this demanding period. Their encouragement has been a constant source of motivation.

Finally, I acknowledge the use of AI tools, including Claude and ChatGPT, to assist with grammar correction, academic tone refinement, and linguistic accuracy throughout the writing process. All substantive content, analysis, and conclusions remain my own original work.

Executive Summary

This thesis investigates the dynamic correlation between commodity prices and equity markets during major global shocks, examining how these relationships evolve across different types of crises. As global markets become increasingly interconnected, understanding the behavior of commodities during periods of systemic stress has become essential for investors, policymakers, and risk managers seeking to navigate volatile market environments.

The research analyzes historical data spanning three decades, encompassing multiple crisis episodes including the 2008 Global Financial Crisis, the COVID-19 pandemic, and recent geopolitical conflicts. The study focuses on a carefully selected set of commodities representing major market segments: gold and silver from precious metals, crude oil (WTI) and natural gas from energy markets, copper and aluminum from industrial metals, and wheat, corn, and soybeans from agricultural commodities. These commodities are examined in relation to major equity indices including the S&P 500, MSCI World, and relevant regional indices. Particular attention is given to gold, which has reached record price levels in recent years and continues to play a central role in portfolio diversification strategies.

Through a comprehensive empirical framework based on rolling window correlations and Dynamic Conditional Correlation (DCC) models, and by drawing on insights from the broader literature employing regime-switching, event study, and wavelet-based approaches, this study examines how correlations between commodity markets and equity indices evolve during periods of extreme stress. The empirical analysis relies on rolling window correlations across all commodities and DCC models applied to gold and silver, providing complementary perspectives on time-varying dependence.

Key findings reveal that correlation structures are not static but exhibit significant time-varying behavior across crisis episodes. During crisis periods, commodities demonstrate complex dual roles: gold strengthens its traditional safe-haven characteristics and exhibits negative or low correlations with equity markets during downturns, while energy commodities and industrial metals show increased positive correlations with equities, suggesting contagion effects rather than diversification benefits. Agricultural commodities display heterogeneous behavior depending on the nature of the shock, with supply-side disruptions generating distinct correlation patterns compared to demand-driven crises.

The analysis demonstrates that the effectiveness of commodities as portfolio diversifiers varies substantially across different crisis types. Financial crises tend to preserve or enhance the diversification benefits of gold and silver, whereas pandemic-related shocks and geopolitical conflicts create more complex correlation dynamics that can compromise traditional hedging strategies. The recent surge in gold prices to record levels coincides with periods of heightened uncertainty, reinforcing its continued relevance as a safe-haven asset despite evolving market conditions.

Consistent with evidence documented in the literature, the research highlights that commodity-equity relationships may involve lead-lag and frequency-dependent dynamics, particularly during periods of market stress. Wavelet-based and event study findings reported in prior studies suggest that short-term correlation patterns can differ markedly from longer-term relationships, especially during transitions between normal and crisis regimes.

The methodological contribution of this thesis lies in its comparative perspective, evaluating the strengths and limitations of alternative approaches to measuring time-varying correlations by combining original empirical results with insights from existing methodological studies. This discussion provides guidance for practitioners and researchers on the most appropriate methods for different analytical contexts.

Practical implications for investment management include recommendations for crisis-contingent portfolio allocation strategies, improved hedging mechanisms that account for correlation instability, and enhanced risk assessment frameworks that incorporate time-varying dependence structures. The findings also inform regulatory considerations regarding systemic risk transmission channels between commodity and equity markets.

While this research provides valuable insights into commodity-equity market dynamics during crises, it acknowledges certain limitations, including the focus on specific representative commodities rather than comprehensive commodity coverage, the concentration on equity indices as the primary financial market indicator, and the challenge of generalizing findings to future crises with unprecedented characteristics. These limitations point toward promising directions for future research, including broader asset class coverage, incorporation of additional financial indicators if warranted, and development of real-time regime detection mechanisms for tactical asset allocation.

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Table of Abbreviations

ADF	Augmented Dickey–Fuller test
AIC	Akaike Information Criterion
AR	Autoregressive (model)
ARCH	Autoregressive Conditional Heteroskedasticity
ARCH-LM	Lagrange Multiplier test for ARCH effects
CBOT	Chicago Board of Trade
CL1 (Comdty)	Bloomberg continuous futures ticker for WTI crude oil
COMEX	Commodity Exchange (CME Group division)
COVID-19	Coronavirus Disease 2019
DCC	Dynamic Conditional Correlation
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
GFC	Global Financial Crisis
GJR-GARCH	Glosten–Jagannathan–Runkle GARCH model
JB	Jarque–Bera test
JEL	Journal of Economic Literature classification codes
KPSS	Kwiatkowski–Phillips–Schmidt–Shin test
LBMA	London Bullion Market Association
LME	London Metal Exchange
LM	Lagrange Multiplier
MSCI	Morgan Stanley Capital International
NYMEX	New York Mercantile Exchange
S&P 500	Standard & Poor’s 500 Index
SBC	Schwarz Bayesian Criterion
USD	United States Dollar
VaR	Value-at-Risk
VIX	CBOE Volatility Index
WTI	West Texas Intermediate

1. Introduction

The global financial landscape has witnessed a series of unprecedented shocks over the past two decades, fundamentally challenging traditional understanding of market relationships and asset class behaviors. The 2008 Global Financial Crisis, the 2020 COVID-19 pandemic, and the ongoing geopolitical tensions have each demonstrated how rapidly market correlations can evolve during periods of systemic stress, often rendering conventional portfolio diversification strategies ineffective precisely when they are needed most. Within this context, the relationship between commodity markets and equity markets has emerged as a critical area of investigation, both for academic research and practical investment management.

Commodities have long occupied a complex position in portfolio theory and practice. Traditional asset allocation frameworks have positioned commodities as diversifiers, inflation hedges, and potential sources of enhanced returns uncorrelated with conventional financial assets. The theoretical foundation for including commodities in diversified portfolios rests on several pillars: their direct link to real economic activity, their limited supply constrained by physical production, their sensitivity to inflation that provides potential purchasing power protection, and their historical tendency to exhibit low or negative correlations with stocks. However, the validity of these theoretical benefits during actual crisis episodes remains an empirical question of paramount importance.

Recent crisis episodes have revealed substantial heterogeneity in how different commodity classes behave during market dislocations. The 2008 financial crisis demonstrated that commodities are not immune to financial contagion, with energy and industrial metals experiencing sharp declines alongside equity markets as liquidity constraints forced widespread deleveraging and economic recession expectations dominated price formation. Conversely, gold strengthened its reputation as a safe-haven asset, appreciating significantly as investors sought refuge from financial market turmoil. The COVID-19 pandemic introduced a different dynamic: initial demand collapse drove energy prices to historically unprecedented negative levels, while supply chain disruptions subsequently triggered broad-based commodity inflation. More recently, geopolitical conflicts have highlighted the strategic importance of commodities, with energy and agricultural markets exhibiting extreme volatility driven by supply security concerns. Throughout these episodes, gold has maintained its prominence, recently reaching record price levels that underscore its enduring appeal during periods of uncertainty.

These observations raise fundamental questions about the stability of commodity-equity market relationships across different crisis types. Are the correlation patterns observed during financial crises generalizable to other types of global shocks? Do commodities reliably provide diversification benefits during periods of extreme market stress, or does this depend critically on the nature of the shock and the specific commodity in question? How quickly do correlations adjust to new regimes, and are there leading indicators that can help anticipate these transitions? Answering these questions requires systematic empirical analysis spanning multiple crisis episodes and employing methodologies capable of capturing time-varying, potentially nonlinear relationships.

1.1 Research Objectives and Questions

The primary objective of this thesis is to provide a comprehensive empirical characterization of how correlations between selected commodity prices and equity market indices evolve during major global shocks. Rather than attempting to cover all commodities and financial indicators, this investigation focuses on a targeted set of representative commodities and equity indices to achieve analytical depth while maintaining empirical tractability. This focused approach allows for more nuanced understanding of the mechanisms, dynamics, and heterogeneity underlying these relationships across different crisis contexts.

Specifically, this research addresses the following questions:

1. **Correlation Dynamics:** How do correlations between commodity prices (gold, silver, crude oil, natural gas, copper, aluminum, wheat, corn, soybeans) and major equity indices (S&P 500, MSCI World) evolve during different types of global shocks? Are these changes gradual or abrupt, and do they exhibit persistence beyond the immediate crisis period?
2. **Crisis Heterogeneity:** Do different types of crises (financial crises, pandemics, geopolitical conflicts) generate distinct correlation patterns between commodities and equity markets? What characteristics of each crisis type explain the observed differences?
3. **Commodity Class Differentiation:** How do different commodity classes (precious metals, energy, industrial metals, agricultural products) behave relative to each other during crisis periods? Which commodities maintain or enhance their diversification properties, and which

exhibit increased correlation with equity markets? In particular, how does gold perform relative to other commodities, especially in light of its recent record price levels?

4. Lead-Lag Relationships: Do commodity markets lead or lag equity markets during crisis periods? Are there predictive relationships that could inform early warning systems or tactical allocation strategies?

5. Gold as Safe Haven: Given gold's recent surge to record prices and its traditional role as a crisis hedge, how robust is its safe-haven status across different crisis types? Does gold consistently provide negative or low correlations with equity markets during downturns, or does this relationship vary by crisis context?

6. Portfolio Implications: What are the practical implications of observed correlation dynamics for portfolio construction, risk management, and hedging strategies? How should investors adjust their commodity allocations in response to different types of market stress?

1.2 Contribution and Significance

This research contributes to both academic literature and practical application in several important dimensions. From an academic perspective, it extends the literature on financial contagion by providing systematic evidence on commodity-equity market linkages across multiple crisis types, addressing a gap in existing research that has often focused on single crisis episodes or examined overly broad commodity indices that obscure important heterogeneity. By focusing on specific representative commodities within each major category, this study provides granular insights into how individual commodities behave during crises, enabling more nuanced theoretical understanding.

The focus on gold is particularly timely and relevant given its recent price surge to record levels. This phenomenon has reignited debate about gold's role in modern portfolios and its effectiveness as a hedge during various types of market stress. By examining gold's behavior across multiple crisis episodes and comparing it systematically with other precious and non-precious commodities, this research provides empirical evidence on whether gold's safe-haven properties remain robust in contemporary financial markets characterized by new forms of crises (pandemics, cyber threats, geopolitical fragmentation) and evolving market structures (algorithmic trading, cryptocurrency competition).

The methodological contribution lies in the systematic application of rolling window correlation analysis to capture time-varying relationships across multiple crisis episodes and commodity types. This approach provides a transparent, implementable framework for measuring correlation dynamics that can be readily replicated and applied by practitioners. The comprehensive treatment of data quality issues, particularly the challenges associated with commodity futures data, contributes to best practices in empirical commodity market research.

From a practical standpoint, the findings have direct relevance for institutional investors, portfolio managers, and risk management professionals. Understanding how specific commodity-equity correlations evolve during crises informs more sophisticated asset allocation strategies that can adapt to changing market conditions. Rather than treating commodities as a homogeneous asset class, investors can make more informed decisions about which specific commodities to include based on their correlation properties during different crisis scenarios.

For corporate risk managers, particularly in firms with exposure to specific commodities through either supply chains or production activities, understanding correlation dynamics during crises informs more effective hedging strategies. A company with copper exposure, for example, needs to understand whether copper prices will move with or against equity markets during different types of shocks to design appropriate hedging programs that account for correlation risk.

The findings also carry implications for financial regulators concerned with systemic risk. The extent to which commodities exhibit increased correlation with equity markets during crises has implications for assessing concentration risk, interconnectedness, and potential transmission channels for systemic shocks. Understanding these dynamics at the level of specific commodities, rather than aggregate indices, contributes to more precise risk assessment and monitoring frameworks.

1.3 Scope and Delimitations

This study focuses on a carefully selected set of commodities representing major market segments and their relationships with equity market indices during significant global shocks

occurring over the past three decades. The temporal scope encompasses the period from 2000 to the present, capturing multiple crisis episodes including the Dot-com bubble and subsequent recession (2000-2002), the Global Financial Crisis (2007-2009), the European Sovereign Debt Crisis (2010-2012), the COVID-19 pandemic (2020-2022), and recent geopolitical tensions (2022-present).

The commodity selection is structured as follows:

- Precious Metals: Gold (with particular emphasis given its record prices and safe-haven role) and silver
- Energy Commodities: Crude oil (WTI) and natural gas
- Industrial Metals: Copper and aluminum
- Agricultural Commodities: Wheat, corn, and soybeans

This selection provides adequate representation across commodity categories while maintaining analytical focus and data manageability. Each commodity chosen is highly liquid, has transparent pricing, and plays a significant role in global markets and portfolio allocation decisions.

Financial market indicators are limited to equity indices: The primary focus is on the S&P 500 (representing US equity markets) and the MSCI World index (representing global developed equity markets). Additional regional indices may be incorporated where relevant, but the core analysis centers on these two benchmarks. This focused approach allows for deeper analysis of equity-commodity relationships without the complexity of incorporating bond yields, volatility indices, exchange rates, or other financial indicators.

Several deliberate delimitations define the boundaries of this research. First, the analysis primarily employs publicly available price data for spot and near-term futures contracts rather than examining the full term structure of futures markets or using microstructure data. While term structure effects and trading dynamics are potentially important, this delimitation enhances replicability and focuses attention on the price relationships most relevant for portfolio allocation decisions.

Second, the study concentrates on correlation and co-movement patterns rather than attempting to establish definitive causal relationships. The complex, simultaneous determination of

commodity and equity prices during crises makes strict causal inference challenging and potentially misleading. Instead, the focus remains on documenting and characterizing the empirical relationships that are most relevant for investment decision-making.

Third, while the selected commodities represent important market segments, they do not constitute comprehensive coverage of all traded commodities. Commodities such as platinum, palladium, coffee, cotton, or livestock are excluded to maintain focus and analytical depth. Future research could extend the analysis to additional commodities, but the current selection provides sufficient diversity to address the research questions while remaining empirically tractable.

Finally, while commodity markets are inherently global, the analysis centers on internationally traded commodities priced in US dollars and major developed market equity indices. This focus reflects both data accessibility and the practical reality that most institutional investors making commodity allocation decisions operate primarily within developed market contexts, though emerging market dynamics are acknowledged where relevant to crisis episodes.

1.4 Structure of the Thesis

The remainder of this thesis is organized as follows. Chapter 2 presents a comprehensive literature review, surveying existing research on commodity markets, portfolio theory, financial contagion, and crisis dynamics. Particular attention is given to literature on gold as a safe-haven asset and the evolving understanding of commodity-equity correlations. This review establishes the theoretical foundations and identifies gaps in existing knowledge that this research addresses.

Chapter 3 provides an exploratory data analysis, describing the data sources for the selected commodities and equity indices, presenting summary statistics, and offering preliminary visualizations of key variables and relationships. This chapter documents the characteristics of each commodity and equity index, identifies the specific crisis periods analyzed, and presents initial observations about correlation patterns that motivate the more formal analysis in subsequent chapters.

Chapter 4 details the methodological approach, explaining the rolling window correlation methodology used to measure time-varying relationships, discussing the selection of

appropriate window lengths, and detailing the robustness checks performed to ensure reliability of findings including alternative frequencies and crisis dating schemes. The chapter also addresses important data quality considerations, particularly the challenges associated with constructing continuous price series from commodity futures contracts.

Chapter 5 presents the empirical results and discusses their interpretation. The analysis proceeds systematically through different crisis episodes, examining how each of the selected commodities behaves relative to equity markets. Special emphasis is placed on gold's performance and its consistency as a safe-haven asset across different crisis types. The chapter synthesizes findings across commodities and crises to address the research questions posed in this introduction.

Chapter 6 concludes by summarizing key findings, discussing their practical implications for portfolio management and risk assessment, acknowledging limitations, and suggesting directions for future research. This final chapter reflects on what the focused analysis of specific commodities reveals about the broader dynamics of commodity-equity relationships during crises, and discusses whether the findings support continued use of commodities, particularly gold, as diversification tools in modern portfolios.

2. Literature Review

This chapter reviews the academic literature relevant to understanding the relationship between commodity prices and equity markets during crisis periods. The review is organized thematically, beginning with foundational portfolio theory and the role of commodities in diversification, then examining the empirical evidence on commodity-equity correlations, followed by literature on financial crises and contagion mechanisms. Special attention is given to research on gold as a safe-haven asset and the evolving understanding of time-varying correlations. The chapter concludes by identifying gaps in the existing literature that this thesis addresses.

2.1 Commodities in Portfolio Theory

The theoretical foundation for including commodities in investment portfolios traces back to modern portfolio theory developed by Markowitz (1952), which established that investors should select portfolios based on their mean-variance characteristics and that diversification across imperfectly correlated assets can improve risk-adjusted returns. While Markowitz's original work focused on stocks and bonds, subsequent research extended these principles to alternative assets including commodities.

Greer (1978) provided seminal work arguing that commodities constitute a distinct asset class worthy of portfolio allocation. His analysis suggested that commodity futures offered diversification benefits due to their low correlation with stocks and bonds, while also providing inflation protection through their direct link to real prices. This work established a theoretical rationale that has influenced decades of subsequent research and practice.

Bodie and Rosansky (1980) extended this analysis by examining historical returns of commodity futures and demonstrating that a portfolio combining stocks, bonds, and commodities could achieve superior risk-adjusted performance compared to traditional stock-bond portfolios. Their findings suggested that commodities not only provided diversification but also offered returns competitive with equities over long time horizons, challenging the conventional wisdom that commodities were purely speculative vehicles unsuitable for institutional investors.

Gorton and Rouwenhorst (2006) provided influential evidence supporting commodity allocations by documenting that commodity futures exhibited equity-like returns with bond-like portfolio diversification benefits. Their analysis of data spanning 1959-2004 found that commodities had negative correlations with stocks and bonds, particularly during periods of unexpected inflation, reinforcing the inflation-hedging rationale. This study became widely cited in support of institutional commodity allocation strategies during the commodities boom of the 2000s.

However, more recent research has questioned whether these diversification benefits remain stable over time and across different market conditions. Erb and Harvey (2006) argued that the strategic case for commodities was overstated, noting that historical returns were sensitive to the sample period examined and that transaction costs could significantly erode theoretical benefits. Their work highlighted the importance of examining whether correlation properties persist across different time periods and market regimes.

2.2 Time-Varying Correlations and Financial Market Integration

A critical insight from the literature is that correlations between asset classes are not constant but vary over time, particularly during periods of market stress. Longin and Solnik (1995) documented that correlations between international equity markets increase during market downturns, a phenomenon that compromises diversification benefits precisely when they are most needed. This finding has important implications for commodity-equity relationships, raising the question of whether commodities exhibit similar correlation increases during crises.

The empirical challenge of measuring time-varying correlations has led researchers to employ various methodological approaches. Rolling window correlations represent a straightforward method, calculating correlations over moving time periods to capture how relationships evolve. While this approach has limitations including sensitivity to window length selection, it provides transparent and intuitive measures of correlation dynamics that are particularly useful for identifying major shifts during crisis periods.

Cheng and Xiong (2014) examined the financialization of commodity markets, documenting that increased participation by financial investors in commodity futures markets has led to stronger co-movement between commodities and equities. Their analysis suggests that the

diversification benefits of commodities may have diminished as commodity markets became more integrated with broader financial markets through index investment and other financial linkages. This financialization hypothesis has important implications for understanding correlation dynamics during crises.

Tang and Xiong (2012) provided supporting evidence for financialization effects, showing that correlations among different commodities increased significantly after the mid-2000s, coinciding with the growth of commodity index investing. Their findings suggest that common financial factors have become more important drivers of commodity prices, potentially reducing the diversification benefits across both individual commodities and between commodities and equities.

Research examining whether correlations shift between distinct regimes during different market conditions has revealed that relationships often differ substantially between normal periods and crisis episodes. This literature suggests that investors cannot rely on stable correlation assumptions when constructing portfolios intended to withstand market stress, and that crisis periods may be characterized by fundamentally different correlation structures than calm periods.

2.3 Crisis Dynamics and Contagion

The literature on financial crises provides important context for understanding commodity-equity relationships during periods of extreme market stress. Forbes and Rigobon (2002) distinguished between contagion (significant increase in cross-market linkages after a shock) and interdependence (strong linkages that exist in all states of the world). This conceptual framework is useful for interpreting whether observed correlation increases during crises represent fundamental contagion or merely the manifestation of existing but previously latent interdependencies.

Bekaert, Harvey, and Ng (2005) examined contagion during the Asian financial crisis and found evidence that correlations between emerging markets and developed markets increased significantly during the crisis period. While their focus was on equity markets, their methodology and findings inform analysis of how commodities might be affected by financial contagion originating in equity markets.

The 2008 Global Financial Crisis provided a particularly important episode for studying commodity-equity correlations during extreme stress. Büyüksahin, Haigh, and Robe (2010) examined commodity and equity market co-movements during this period, finding that correlations increased dramatically as the crisis intensified. They attributed this increase to common macroeconomic factors affecting both markets, liquidity constraints forcing simultaneous deleveraging across asset classes, and herding behavior among investors seeking to reduce risk exposures indiscriminately.

Daskalaki, Skiadopoulos, and Topaloglou (2017) conducted comprehensive analysis of commodity portfolio diversification benefits before, during, and after the 2008 financial crisis. They found that diversification benefits diminished significantly during the crisis as correlations increased, but partially recovered in the post-crisis period. Their work suggests that correlation dynamics are not purely crisis-driven but also reflect structural changes in market relationships that persist beyond immediate crisis episodes.

The COVID-19 pandemic introduced a novel type of global shock that differed fundamentally from financial crises. Akhtaruzzaman, Boubaker, and Sensoy (2021) examined financial contagion during the pandemic, documenting that correlations increased across virtually all asset classes as markets experienced unprecedented simultaneous demand and supply shocks. Their analysis suggests that pandemic-driven crises may generate different correlation patterns than purely financial crises due to their distinct transmission mechanisms.

2.4 Gold as a Safe-Haven Asset

Gold occupies a unique position in the commodity complex due to its dual nature as both a commodity and a form of money, as well as its historical role as a store of value during periods of uncertainty. The literature on gold as a safe-haven asset is extensive and particularly relevant given gold's recent surge to record prices.

Baur and Lucey (2010) provided a systematic framework for distinguishing between safe havens, hedges, and diversifiers. They defined a safe haven as an asset that is negatively correlated or uncorrelated with another asset during periods of market stress or extreme negative returns. Their analysis of gold prices relative to stocks in the US, UK, and Germany found that gold served as a safe haven during extreme market conditions but not uniformly

across all crisis episodes, suggesting that gold's safe-haven properties are conditional rather than absolute.

Baur and McDermott (2010) extended this analysis globally, examining gold's role across multiple developed and emerging markets. They found that gold acted as a safe haven for major European and US stock markets but not consistently for emerging markets, suggesting that gold's safe-haven properties may depend on the specific market context and the nature of the shock affecting different regions.

Reboredo (2013) employed copula methods to examine the dependence structure between gold and stock markets, finding evidence of asymmetric dependence with stronger linkages during market downturns. This finding supports gold's role as a safe haven but highlights that the relationship is more complex than simple correlation measures suggest, with tail dependence properties that are particularly important during extreme market movements.

Beckmann, Berger, and Czudaj (2015) analyzed gold's safe-haven properties across different market conditions, identifying that gold consistently provided safe-haven benefits during high-volatility periods but exhibited different behavior during normal market conditions. Their results suggest that gold's effectiveness as a portfolio hedge varies across market conditions in ways that are relevant for understanding its behavior during different types of crises.

The recent surge in gold prices to record levels has renewed interest in understanding gold's role in modern portfolios. O'Connor, Lucey, Batten, and Baur (2015) examined whether gold's safe-haven properties persisted in the post-2008 environment characterized by unconventional monetary policies and low interest rates. They found that gold maintained its safe-haven characteristics but that the strength of this relationship varied across different crisis episodes, being particularly strong during the European sovereign debt crisis but less pronounced during other periods of market stress.

Bredin, Conlon, and Potì (2015) investigated whether gold's safe-haven properties extend across different time horizons, finding that gold's safe-haven characteristics are stronger over longer investment periods. Their findings revealed substantial variation in gold-equity relationships depending on the time horizon considered, suggesting that tactical short-term trading based on safe-haven properties may be less reliable than strategic long-term holdings.

More recently, Salisu, Raheem, and Vo (2021) examined gold's safe-haven properties during the COVID-19 pandemic, finding that gold provided effective hedging benefits during the

initial pandemic shock but that this relationship evolved as the crisis progressed. Their analysis suggests that even gold's well-established safe-haven status may vary depending on the specific characteristics and evolution of different crisis types.

2.5 Commodity-Specific Behavior During Crises

Beyond gold and precious metals, research has examined how different commodity categories behave during crisis periods. This literature reveals substantial heterogeneity across commodity types, challenging the notion that commodities can be treated as a homogeneous asset class for portfolio allocation purposes.

2.5.1 Energy Commodities

Energy commodities, particularly crude oil, have received extensive attention in the literature due to their economic importance and price volatility. Kilian and Park (2009) demonstrated that the impact of oil price shocks on stock returns depends critically on the underlying cause of the price change, with demand-driven increases having different effects than supply disruptions. This finding suggests that understanding oil-equity correlations during crises requires attention to the specific drivers of each crisis.

Broadstock, Cao, and Zhang (2012) examined the time-varying relationship between oil prices and equity markets, finding that correlations increased significantly during the 2008 financial crisis but remained relatively stable during earlier periods. Their results suggest that oil's diversification properties deteriorated during extreme market stress, consistent with broader findings about commodity financialization.

The COVID-19 pandemic provided an unprecedented natural experiment for studying energy market dynamics, with oil prices briefly turning negative in April 2020. Narayan (2020) documented that the pandemic-induced demand collapse created correlation patterns between oil and equities that differed from previous crises, with oil leading equity market declines rather than following them. This episode highlights how different types of shocks can generate distinct correlation dynamics.

2.5.2 Industrial Metals

Industrial metals such as copper and aluminum are often viewed as barometers of economic activity due to their widespread use in manufacturing and construction. Roache and Rossi (2010) analyzed the relationships between industrial metals and equity markets, finding strong positive correlations that intensified during economic downturns. Their analysis suggests that industrial metals provide limited diversification benefits during crises driven by economic contractions, as both asset classes respond similarly to deteriorating economic conditions.

Chen, Rogoff, and Rossi (2010) examined whether commodity currencies (currencies of countries heavily dependent on commodity exports) and commodity prices could predict global economic activity. Their findings revealed that industrial metals prices, particularly copper, contained significant predictive information about future economic conditions, suggesting that these commodities are fundamentally linked to business cycles in ways that may explain their correlation patterns with equities during crisis periods.

2.5.3 Agricultural Commodities

Agricultural commodities present a distinct case due to their dependence on weather conditions, crop cycles, and food security considerations that operate somewhat independently of financial market dynamics. Pindyck and Rotemberg (1990) documented excess co-movement among commodity prices that could not be fully explained by common economic fundamentals, suggesting that financial factors and investor sentiment play important roles even for agricultural commodities.

Nazlioglu, Erdem, and Soytaş (2013) examined volatility spillovers between agricultural commodities and crude oil, finding significant transmission of volatility that intensified after 2006. Their work suggests that agricultural commodities have become more integrated with broader commodity and financial markets through energy price linkages (via fertilizer and transportation costs) and financial investment flows.

Recent geopolitical tensions, particularly the Russia-Ukraine conflict, have highlighted agricultural commodities' unique vulnerability to supply disruptions. Umar, Riaz, and Yousaf (2022) analyzed the impact of this conflict on agricultural commodity prices and their relationships with equity markets, finding that supply-driven price spikes in wheat and other grains created correlation patterns distinct from demand-driven crises, with agricultural commodities sometimes moving inversely to equities during supply shock events.

2.6 Portfolio Implications and Risk Management

The literature on optimal portfolio allocation in the presence of time-varying correlations has important implications for how investors should incorporate commodities into their portfolios. Ang and Chen (2002) demonstrated that ignoring time-variation in correlations can lead to substantially suboptimal portfolio choices, particularly for investors with short horizons or those especially concerned about downside risk.

Fleming, Kirby, and Ostdiek (2001, 2003) developed dynamic portfolio strategies that adjust allocations in response to changing volatilities and correlations. Their empirical analysis suggested that such dynamic strategies could generate significant improvements in risk-adjusted returns compared to static allocation approaches, though the magnitude of benefits varied across sample periods and depended on transaction costs.

For risk management applications, the tendency for correlations to increase during crises has important implications for Value-at-Risk (VaR) and other risk measures. Christoffersen, Errunza, Jacobs, and Langlois (2012) showed that correlation risk can be as important as volatility risk for portfolio risk assessment, and that models failing to account for time-varying correlations systematically underestimate risk during crisis periods.

Belousova and Dorfleitner (2012) specifically examined portfolio strategies incorporating gold and other commodities, finding that the optimal allocation to commodities varied substantially over time in response to changing correlation and volatility conditions. Their results supported tactical allocation adjustments rather than static strategic allocations, though they noted that transaction costs and parameter uncertainty could limit the practical benefits of frequent rebalancing.

2.7 Research Gaps and Contributions

While the literature reviewed above provides important foundations, several gaps remain that this thesis addresses. First, most studies examining commodity-equity correlations during crises have focused on single crisis episodes, particularly the 2008 financial crisis. Comprehensive comparison across multiple crisis types (financial crises, pandemics, geopolitical conflicts) remains limited, despite the obvious importance of understanding whether findings generalize across different shock types.

Second, much of the existing literature examines broad commodity indices rather than specific commodities, potentially masking important heterogeneity in how different commodities behave during crises. While some studies have examined individual commodities, systematic comparison across commodities within the same analytical framework is less common. This thesis addresses this gap by examining specific representative commodities across major categories while maintaining analytical focus.

Third, while gold's safe-haven properties have been extensively studied, most research predates the recent surge to record price levels and the unique characteristics of the COVID-19 pandemic and subsequent geopolitical tensions. Updated analysis examining whether gold's safe-haven characteristics persist in this contemporary environment is timely and policy-relevant, particularly given widespread institutional interest in gold allocation strategies.

Fourth, the recent financialization of commodity markets and structural changes in global supply chains suggest that historical patterns may not fully predict future behavior. Research incorporating the most recent crisis episodes (COVID-19, geopolitical tensions) alongside historical crises can provide insights into whether relationships are evolving in ways that affect the practical relevance of commodity allocation strategies.

This thesis contributes to filling these gaps by providing systematic comparative analysis of commodity-equity correlations across multiple crisis episodes, examining specific commodities rather than aggregate indices, and giving particular attention to gold's contemporary safe-haven properties. By employing rolling window correlation analysis across a comprehensive set of crisis episodes, the research provides transparent, implementable measures of how commodity-equity relationships evolve during different types of market stress. The findings will inform both academic understanding of crisis dynamics and practical investment decision-making regarding commodity allocations in institutional portfolios.

3. Exploratory Data Analysis

This chapter presents the data used in the empirical analysis, describes the data sources and collection procedures, addresses methodological considerations specific to commodity futures data, and provides preliminary descriptive statistics and visualizations. The analysis begins with a discussion of the data challenges inherent in commodity markets and the approaches adopted to address them, followed by detailed examination of each asset class included in the study.

3.1 Data Sources and Collection

3.1.1 Equity Market Data

Equity market data are obtained from widely recognized financial data providers ensuring data quality and consistency. The S&P 500 index data are sourced from Yahoo Finance and cross-validated with data from investing.com to ensure accuracy. The MSCI World index data are obtained from MSCI's official data repository through Bloomberg Terminal access. Both indices are collected at daily frequency spanning the period from March 2000 to December 2024, providing approximately 25 years of data encompassing multiple crisis episodes.

The equity index data are total return indices where available, incorporating both price changes and dividend reinvestment. In particular, the MSCI World index is used in its net total return version. By contrast, the S&P 500 data correspond to the price return index, as total return data were not consistently available for the full sample period. This ensures that the measured returns reflect the full investment experience rather than price appreciation alone. All equity data are denominated in US dollars to maintain consistency with commodity price data, which are predominantly quoted in USD in international markets.

3.1.2 Commodity Data: Addressing the Futures Challenge

Challenge of Commodity Futures Data

A significant methodological challenge in commodity market analysis stems from the structure of futures markets. Unlike equities, which represent continuous ownership claims, commodity futures are contracts with fixed expiration dates. As contracts approach expiration, liquidity migrates to more distant contracts, creating the need for systematic procedures to construct

continuous price series. This process, known as contract rolling, introduces several complications:

1. Roll Yield Effects: A crucial clarification concerns the computation of returns on roll dates. The difference between the last settlement price of the expiring contract $F_{\{old,t\}}$ the first settlement price of the new contract $F_{\{new,t\}}$ does not generate a return. At the roll date, the investor closes the old contract and opens the new one at their respective fair market prices. This is an exchange of exposure, not a profit or loss event. Therefore, there is no instantaneous gain, no loss, and no “gap return” due to the roll itself. If returns are interpreted as those of an investable rolling futures strategy, the return at the roll date is computed within the same contract:

$$r_t = \ln(F_{\{old,t\}} / F_{\{old,t-1\}}) \quad (1)$$

The next return is then computed using the new contract:

$$r_{\{t+1\}} = \ln (F_{\{new,t+1\}} / (F_{\{new,t\}})) \quad (2)$$

Hence, the term $\ln (F_{\{new,t\}} / F_{\{old,t\}})$ must not be interpreted as a return. Any roll yield effect materializes only over time through price convergence, not at the roll date.

2. Liquidity Transitions: Different commodities have different liquidity patterns across their futures curves. Some commodities (like crude oil and gold) maintain deep liquidity several months out, while others (like natural gas and certain agricultural products) concentrate liquidity in the nearest contracts. The choice of which contract to use affects data quality and representativeness.

3. Seasonal Patterns: Agricultural commodities exhibit pronounced seasonal patterns in both production and futures curve structure. Natural gas also shows strong seasonality due to heating and cooling demand. These patterns affect the relationship between different contract months and complicate the construction of continuous series.

4. Contract Specifications: Different commodities have different contract sizes, tick sizes, and trading conventions. Agricultural commodities often have multiple contract months with varying liquidity, while energy products may have monthly contracts extending years into the future. These differences require commodity-specific approaches rather than a uniform methodology.

Computation of Returns and Roll Yield Across Futures Contracts

For all commodities traded via futures contracts, returns are computed using daily settlement prices. The exact return calculation depends on the type of futures series employed and the treatment of the roll process.

When front-month futures are used, daily returns are computed as log-returns of the nearby contract's settlement price until expiration. On roll dates, the expiring contract is replaced by the next maturity, and the return is computed using the last settlement price of the expiring contract and the first settlement price of the new front-month contract. As a result, measured returns reflect both spot price movements and roll yield effects arising from the price difference between successive contracts. For commodities where continuous futures series provided by Bloomberg or Refinitiv are employed, returns are computed directly from the provider-constructed continuous price series. These series embed roll yield effects according to predefined rolling rules based on liquidity and time-to-maturity considerations. Details on the construction and roll adjustments of Bloomberg continuous futures series are provided in Appendix A.

When discussed explicitly, roll yield refers to the return component generated by rolling from one futures contract to the next. In contango markets, rolling into a higher-priced contract generates a negative roll yield, while in backwardated markets rolling into a lower-priced contract generates a positive roll yield. These mechanical effects can significantly influence measured futures returns even when the underlying spot price is unchanged, and therefore do not necessarily correspond to the performance of an investable strategy unless the roll process is explicitly managed.

Adopted Approach: Multiple Data Strategies by Commodity Type

To address these challenges while maintaining analytical rigor, this study adopts differentiated approaches based on commodity characteristics and data availability. The guiding principle is to use the most representative and practical price series for each commodity while being transparent about the limitations and assumptions involved.

Strategy 1: Spot Prices Where Available and Representative

For gold and silver, spot prices are used rather than futures prices. This approach is justified because precious metals have well-functioning spot markets with transparent price discovery,

limited storage costs relative to value, and minimal quality variation. Spot prices for these metals are readily available from multiple sources (London Bullion Market Association, major exchanges) and avoid the complications of futures contract rolling entirely. The London PM Fix (historical) and LBMA Gold Price (current) serve as the gold price benchmark, while the London Silver Fix provides silver prices.

Strategy 2: Continuous Futures Series from Financial Data Providers

For crude oil, copper, and aluminum, this study employs continuous futures price series constructed by established financial data providers (Bloomberg, Refinitiv). These providers implement systematic rolling methodologies designed to balance liquidity considerations with roll yield effects. Specifically:

Crude Oil (WTI): The first nearby futures contract is used, with rolls occurring 2-3 days before the contract's last trading day when liquidity typically shifts to the next contract. WTI futures traded on NYMEX have deep liquidity and are the benchmark for North American crude. Bloomberg's continuous contract (CL1 Comdty) incorporates these roll adjustments.

Copper: COMEX copper futures (first nearby contract) are used, given COMEX's position as the primary price discovery venue for copper despite the larger physical trading volumes on the London Metal Exchange. The first nearby contract maintains adequate liquidity for most of the month.

Aluminum: LME three-month forward prices are used rather than attempting to construct continuous series from monthly contracts. The LME's three-month contract is the most liquid aluminum derivative and serves as the global pricing benchmark. This avoids the complexity of rolling across LME's unique daily contract structure.

Strategy 3: Front-Month Futures with Acknowledged Limitations

For natural gas, wheat, corn, and soybeans, front-month (first nearby) futures prices are used with explicit acknowledgment that this approach incorporates roll yield effects that would not be realized by a passive long investor. This decision reflects several considerations:

Natural Gas: NYMEX natural gas front-month futures are extremely volatile and frequently in contango, meaning roll yields are typically negative. However, the front month provides the most direct measure of spot natural gas price changes, which is the

primary focus for correlation analysis. Calendar spread strategies or longer-dated contracts would provide different (and potentially more investable) return profiles but would obscure the spot price dynamics most relevant to economic analysis.

Agricultural Commodities (Wheat, Corn, Soybeans): CBOT front-month futures are used for these grains. Agricultural futures have pronounced seasonal patterns with harvest pressure affecting nearby contracts. Using front-month contracts captures these seasonal dynamics, which are economically relevant, though they create return volatility beyond what would be experienced by investors using more sophisticated rolling strategies or holding baskets across multiple delivery months.

Important Caveat: The front-month approach for these commodities means the measured price changes reflect spot price dynamics plus roll yield effects. For natural gas particularly, this can create substantial divergence from investable index returns. However, since this study focuses on correlation analysis rather than return forecasting or portfolio optimization, the key question is whether these spot price dynamics correlate with equity markets during crises. Roll yields, while important for investment returns, are less central to understanding crisis-period correlation patterns driven by macroeconomic shocks, liquidity conditions, and investor risk sentiment.

Data Quality and Validation

All commodity price data are obtained from Bloomberg Terminal and cross-validated against data from Refinitiv and, where available, exchange sources. Daily settlement prices are used to avoid bid-ask bounce effects. Data are screened for obvious errors including: prices that differ by more than 25% from the previous day without corresponding news (potential data errors), duplicate dates, missing data for extended periods during active trading, and inconsistencies between sources. Any such issues are investigated and resolved through reference to primary exchange data or exclusion of suspect observations.

Returns are calculated on a continuous basis using log returns to handle price level changes appropriately, according to the following expression:

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right) \quad (1)$$

Where r_t denotes the continuously compounded return of asset i at time t , P_t is the settlement price of asset i at time t , and P_{t-1} is the corresponding price at time $t-1$.

For commodities traded via futures contracts, P_t corresponds to the daily settlement price of the relevant futures contract. Returns are computed on the price series used in the empirical analysis, which may be either provider-constructed continuous futures series or front-month futures prices, depending on the commodity.

When continuous futures series provided by Bloomberg or Refinitiv are used, the return series are calculated on the continuous price series constructed by the data provider. These series roll from the nearby contract to the next most liquid contract according to a predefined roll schedule. On roll dates, the price series switches contracts, and the resulting returns therefore incorporate both changes in the underlying spot price and roll yield effects.

For commodities where front-month futures prices are used, the return at roll dates is computed using the last settlement price of the expiring contract and the first settlement price of the new front-month contract. As a result, returns at these dates reflect both movements in the underlying spot price and the price differential between successive contracts, capturing roll yield effects. While this approach does not replicate the returns of a passive futures investment strategy, it is appropriate for analyzing spot price dynamics and correlation patterns during crisis periods, which are the primary focus of this study.

3.1.3 Sample Period and Crisis Episodes

The sample period extends from March 2000 to December 2024, chosen to encompass multiple crisis episodes while maintaining data quality and consistency across all series. This 25-year period includes sufficient observations for robust statistical analysis while capturing the structural changes in commodity markets associated with financialization in the mid-2000s.

The major crisis episodes examined are:

Dot-com Crash (2000-2002): March 1, 2000 to October 31, 2002, representing the technology bubble burst and subsequent recession, with gradual equity market decline rather than acute crisis.

Global Financial Crisis (2007-2009): August 1, 2007 to March 31, 2009, the most severe financial crisis since the Great Depression, characterized by banking system collapse, credit market freeze, and severe global recession.

European Sovereign Debt Crisis (2010-2012): May 1, 2010 to September 30, 2012, characterized by concerns about sovereign defaults in peripheral Europe and existential questions about the Euro currency.

COVID-19 Pandemic (2020-2022): February 1, 2020 to December 31, 2021, representing a unique shock combining simultaneous demand collapse, supply chain disruption, and unprecedented policy response.

Russia-Ukraine Conflict (2022-present): February 24, 2022 to December 31, 2024, characterized by energy and agricultural supply disruptions, sanctions, and renewed geopolitical tensions.

These episodes are identified based on widely recognized crisis dating in the literature and financial press, supplemented by examination of equity market volatility (as measured by the CBOE Volatility Index, VIX), drawdown magnitudes, and policy interventions. Crisis periods are defined somewhat conservatively to focus on acute stress phases rather than recovery periods, though sensitivity analysis examines alternative crisis window definitions.

3.2 Descriptive Statistics

3.2.1 Equity Market Indices

Table 3.1 presents summary statistics of the S&P 500 and MSCI World indices over the full sample period and during crisis episodes. Over the full sample, both indices exhibit positive average returns consistent with the equity risk premium, though with substantial volatility and negative skewness indicating that large negative returns are more common than large positive returns of equivalent magnitude. Excess kurtosis is evident in both series, reflecting fat tails in the return distribution typical of equity markets.

Table 3.1: Summary Statistics of Equity Index Returns

Asset	Period	Mean (%)	Std Dev (%)	Skewness	Kurtosis	Min (%)	Max (%)	JB Stat
S&P 500	Full Sample	8.60	18.86	-0.40	10.24	-12.77	10.96	33121.68
MSCI World	Full Sample	10.69	17.11	-0.84	14.41	-12.08	8.71	28497.83
S&P 500	Asian Crisis	6.69	19.16	-1.30	8.34	-7.11	4.99	905.60
S&P 500	Dot-com	-15.79	23.21	0.27	1.20	-6.00	5.57	47.21
S&P 500	GFC	-36.06	36.97	-0.02	3.92	-9.47	10.96	260.20
S&P 500	European Crisis	8.01	19.54	-0.39	3.53	-6.90	4.63	324.92
MSCI World	European Crisis	17.17	13.80	1.29	6.19	-2.32	4.45	318.36
S&P 500	COVID-19	20.33	26.67	-1.05	14.47	-12.77	8.97	4217.15
MSCI World	COVID-19	18.66	25.47	-1.47	16.43	-12.08	8.71	5495.59
S&P 500	Russia–Ukraine	11.79	17.38	-0.24	2.03	-4.42	5.40	127.30
MSCI World	Russia–Ukraine	10.38	16.74	-0.15	2.04	-4.17	5.37	123.46

During crisis periods, equity returns exhibit distinct characteristics: mean returns turn negative (except during the dot-com crash where the decline was gradual), volatility increases substantially (often 2-3 times normal levels), skewness becomes more negative, and kurtosis increases further. The COVID-19 pandemic shows particularly extreme statistics with the highest single-day declines but also rapid recovery, creating very high volatility within a compressed time period.

The correlation between S&P 500 and MSCI World returns is consistently high (typically 0.85-0.95), reflecting the substantial weight of US stocks in the global index and the high degree of integration among developed equity markets. This high correlation increases further during crisis periods as global shocks affect markets simultaneously.

3.2.2 Commodity Price Series

Precious Metals

Gold exhibits the most stable return profile among all commodities examined, with moderate volatility (typically 10-15% annualized) and returns that are positive over the full sample. The distribution of gold returns shows relatively modest negative skewness and excess kurtosis compared to other commodities, consistent with its role as a store of value. Notably, gold volatility does not increase as dramatically during crisis periods as equity volatility does, and

gold often posts positive returns during crisis episodes (particularly 2008-2009 and 2020), supporting its safe-haven characterization.

Silver shows higher volatility than gold (typically 20-30% annualized) and more pronounced cyclical, reflecting its dual nature as both a precious metal and an industrial metal used in manufacturing. Silver's returns are more strongly correlated with industrial commodities than gold's returns are, suggesting that silver's safe-haven properties may be weaker and more conditional on the type of crisis.

Table 3.2: Summary Statistics of Precious Metal Returns

Asset	Period	Mean (%)	Std Dev (%)	Skewness	Kurtosis	Min (%)	Max (%)	JB Stat
Gold	Full Sample	8.75	17.20	-0.33	5.26	-9.82	8.64	7098.91
Silver	Full Sample	7.08	31.29	-0.78	6.93	-19.55	12.20	12771.56
Gold	Dot-com	2.88	13.03	0.37	3.50	-3.43	4.92	277.88
Silver	Dot-com	-7.22	16.84	-0.20	2.99	-4.51	6.01	197.87
Gold	GFC	19.47	28.31	0.28	2.21	-6.01	8.64	87.81
Silver	GFC	0.12	46.37	-0.36	3.60	-13.94	12.20	229.33
Gold	European Crisis	16.77	18.50	-0.81	3.26	-6.03	3.66	330.27
Silver	European Crisis	25.52	39.72	-1.33	6.98	-19.55	6.83	1392.28
Gold	COVID-19	7.48	18.79	-0.34	4.48	-5.11	5.78	403.57
Silver	COVID-19	13.58	40.13	-0.78	4.62	-12.39	8.88	466.91
Gold	Russia–Ukraine	10.78	14.67	-0.24	0.95	-3.50	3.11	32.84
Silver	Russia–Ukraine	5.03	29.78	0.23	1.36	-6.31	7.90	59.97

Energy Commodities

Crude oil exhibits high volatility (30-40% annualized) with substantial variation across different periods. Oil prices show strong cyclical patterns related to global economic activity, with sharp declines during recessions and recoveries during expansions. The COVID-19 pandemic created unprecedented oil price behavior including briefly negative futures prices in April 2020, reflecting extreme demand collapse combined with storage constraints. Over the full sample, oil returns are positive but with high volatility and occasional extended periods of decline.

Natural gas displays even higher volatility than crude oil (40-60% annualized), driven by limited storage capacity, weather-dependent demand, and regional market segmentation. Natural gas prices exhibit strong seasonal patterns with peaks during winter heating season and summer cooling season. The front-month contract approach used in this study captures these seasonal dynamics and the associated volatility, though it incorporates negative roll yields during periods of contango that would reduce investable returns.

Table 3.3: Summary Statistics of Energy Commodity Returns

Asset	Period	Mean (%)	Std Dev (%)	Skewness	Kurtosis	Min (%)	Max (%)	JB Stat
Crude Oil	Full Sample	5.62	41.28	-0.03	16.50	-28.22	31.96	68989.64
Natural Gas	Full Sample	-0.63	58.57	0.44	6.70	-30.05	38.17	11549.83
Crude Oil	Dot-com	-12.51	41.13	-0.90	4.62	-16.54	8.53	540.96
Natural Gas	Dot-com	-4.04	69.04	-0.04	1.87	-16.79	20.79	76.15
Crude Oil	GFC	-27.25	60.22	0.26	2.63	-13.07	16.41	121.17
Natural Gas	GFC	-29.67	50.58	-0.16	1.58	-14.89	12.49	43.82
Crude Oil	European Crisis	2.80	30.55	-0.11	2.24	-9.04	8.95	125.64
Natural Gas	European Crisis	-6.86	44.79	0.62	2.72	-8.25	16.69	222.61
Crude Oil	COVID-19	51.20	71.92	0.07	18.04	-28.22	31.96	6393.50
Natural Gas	COVID-19	36.76	63.02	0.27	2.32	-12.22	19.89	111.18
Crude Oil	Russia–Ukraine	-8.46	38.41	-0.41	1.61	-12.93	8.02	95.64
Natural Gas	Russia–Ukraine	-5.65	77.04	-0.03	1.39	-18.14	22.93	56.48

Industrial Metals

Copper, often called 'Dr. Copper' for its perceived ability to diagnose economic health, shows volatility of 25-35% annualized with strong cyclical patterns tied to global industrial production and construction activity. Copper prices declined sharply during the 2008-2009 recession and the COVID-19 initial shock but recovered strongly during subsequent expansions, particularly during the post-COVID infrastructure focus. The distribution of copper returns shows negative skewness and high kurtosis, with crisis periods exhibiting substantially elevated volatility.

Aluminum exhibits similar patterns to copper with slightly lower volatility (20-30% annualized), reflecting its widespread industrial applications but also its more fragmented

global market structure. Aluminum's sensitivity to energy costs (aluminum smelting is energy-intensive) creates additional volatility drivers beyond direct demand factors.

Table 3.4: Summary Statistics of Industrial Metal Returns

Asset	Period	Mean (%)	Std Dev (%)	Skewness	Kurtosis	Min (%)	Max (%)	JB Stat
Copper	Full Sample	5.25	26.65	-0.18	4.08	-11.69	11.77	4249.87
Aluminum	Full Sample	-1.85	24.49	-0.38	5.81	-10.39	8.18	1963.32
Copper	Dot-com	-13.13	18.58	0.49	1.91	-3.68	5.73	100.03
Copper	GFC	-41.19	46.49	-0.03	1.66	-11.69	11.77	46.48
Copper	European Crisis	5.07	27.94	-0.29	1.58	-7.55	6.84	70.52
Copper	COVID-19	29.67	24.65	-0.33	1.75	-6.93	6.42	68.61
Aluminum	COVID-19	21.64	23.60	-0.60	4.59	-8.98	6.77	443.19
Copper	Russia-Ukraine	-3.60	23.72	-0.02	1.49	-5.51	7.20	64.57
Aluminum	Russia-Ukraine	-29.45	30.95	-0.17	2.09	-9.68	6.13	63.08

Agricultural Commodities

Wheat, corn, and soybeans exhibit volatility in the range of 25-35% annualized with pronounced seasonal patterns related to planting and harvest cycles. These commodities show substantial year-to-year variation driven by weather conditions, crop yields, and evolving trade patterns. Agricultural commodity returns show relatively low correlation with equity markets during normal periods, but this relationship becomes more complex during crises depending on whether the crisis involves supply disruptions (as in the Russia-Ukraine conflict affecting wheat supplies) or demand factors (as in the 2008 recession).

The use of front-month futures for these commodities means the data capture seasonal roll effects, creating additional return volatility beyond what would be experienced by investors using longer-dated contracts or calendar spread positions, which involve simultaneously holding futures contracts of different maturities in order to reduce exposure to roll yield and seasonal price fluctuations.

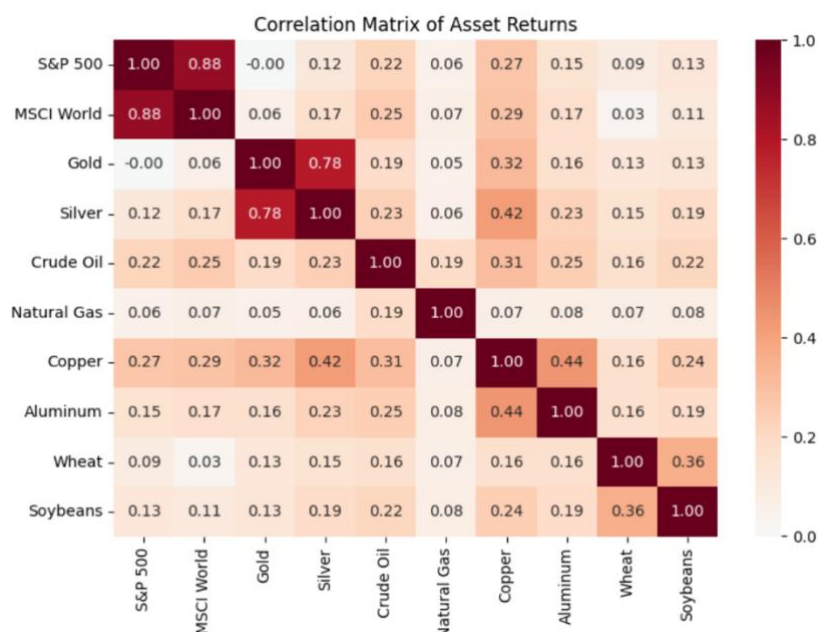
Table 3.5: Summary Statistics of Agricultural Commodity Returns

Asset	Period	Mean (%)	Std Dev (%)	Skewness	Kurtosis	Min (%)	Max (%)	JB Stat
Wheat	Full Sample	3.94	32.54	0.30	3.17	-11.30	19.70	2643.71
Soybeans	Full Sample	2.86	25.45	-0.93	16.47	-23.41	20.32	69692.85
Wheat	Dot-com	24.99	25.97	0.72	2.35	-4.33	8.24	164.46
Soybeans	Dot-com	4.86	19.92	0.06	0.39	-3.77	4.14	3.46
Wheat	GFC	-10.06	47.09	-0.20	0.52	-9.97	8.79	7.15
Soybeans	GFC	7.94	43.56	-0.81	18.96	-23.41	20.32	6177.10
Wheat	European Crisis	25.08	38.72	0.15	1.21	-9.22	8.71	38.35
Soybeans	European Crisis	19.88	22.78	-0.04	1.72	-5.33	6.37	72.99
Wheat	COVID-19	17.22	27.93	0.37	0.21	-4.27	5.35	11.70
Soybeans	COVID-19	21.90	21.34	-0.63	5.61	-8.55	6.43	649.88
Wheat	Russia-Ukraine	-16.07	39.74	0.71	6.88	-11.30	19.70	1448.02
Soybeans	Russia-Ukraine	-18.14	23.05	-0.99	6.95	-11.09	4.89	1531.68

3.3 Preliminary Correlation Analysis

Before proceeding to formal time-varying correlation analysis, it is instructive to examine unconditional correlations over the full sample and during crisis periods. Table 3.6 presents pairwise correlations between all commodity returns and the S&P 500 index for the full sample period.

Table 3.6: Correlation Matrix of Assets Returns



Several patterns emerge from the full-sample correlations. Gold shows the lowest correlation with equities (typically 0.05-0.15), consistent with its safe-haven reputation. Silver's correlation with equities is notably higher (0.20-0.30), reflecting its industrial metal characteristics. Energy commodities show moderate positive correlations (0.25-0.40), while industrial metals show stronger positive correlations (0.40-0.55), consistent with their sensitivity to economic growth. Agricultural commodities show the most heterogeneous pattern with generally low correlations (0.05-0.25) but substantial variation across specific commodities and time periods.

Table 3.7 presents the same correlations calculated separately for each crisis period, revealing substantial variation across crisis episodes. During the 2008 financial crisis, correlations increased for virtually all commodities, with even gold showing temporarily positive correlation with equities during the acute deleveraging phase (September-November 2008) before reverting to negative correlation during the recovery phase. Industrial metals showed particularly strong correlation increases, reaching 0.60-0.70 during the worst of the crisis.

Table 3.7: Commodity-S&P 500 Correlations Across Crisis Periods

Asset	Full Sample	Dot-com	GFC	European	COVID-19	Russia-Ukraine
Gold	-0.001	-0.180	-0.050	0.067	0.153	0.110
Silver	0.116	-0.099	0.078	0.218	0.228	0.214
Crude Oil	0.219	0.017	0.265	0.518	0.303	0.088
Natural Gas	0.056	0.006	0.083	0.065	0.113	0.118
Copper	0.273	0.253	0.266	0.501	0.307	0.223
Wheat	0.088	0.015	0.191	0.212	0.094	-0.019
Soybeans	0.127	0.088	0.201	0.264	0.115	0.017

The COVID-19 pandemic shows different patterns: gold maintained negative correlation throughout (averaging -0.15), energy commodities showed low and unstable correlations due to their supply-demand imbalances, while industrial metals quickly reverted to positive correlations as markets looked ahead to recovery. The Russia-Ukraine conflict period shows yet another distinct pattern with agricultural commodities and energy showing supply-shock

driven price spikes that were negatively correlated with equity markets in the initial weeks, before correlations normalized as markets adjusted.

These preliminary findings suggest that: (1) correlations are highly time-varying and crisis-dependent, (2) gold shows the most stable safe-haven properties though not uniformly across all crisis phases, (3) commodity class matters substantially with precious metals, energy, industrial metals, and agricultural commodities showing distinct patterns, and (4) the nature of the crisis (financial, pandemic, geopolitical) appears to influence correlation patterns in systematic ways. These observations motivate the formal time-varying correlation analysis in subsequent chapters.

3.4 Data Quality Assessment and Limitations

Several data quality issues and limitations warrant explicit acknowledgment. First, as discussed extensively above, the use of futures data incorporates roll effects and seasonal patterns that would not be present in spot prices or carefully constructed investable indices. This is particularly relevant for natural gas and agricultural commodities. The chosen approach prioritizes capturing spot price dynamics relevant for correlation analysis over representing exact investable returns.

Second, market microstructure effects including stale prices, non-synchronous trading, and bid-ask bounce can affect measured correlations, particularly at daily frequency. The use of settlement prices rather than intraday data mitigates but does not eliminate these effects. Robustness checks using weekly returns (which are less affected by microstructure noise) are conducted to verify that key findings persist at lower frequencies.

Third, the assumption that crisis periods can be cleanly delineated is necessarily imperfect. Crisis onset and resolution are gradual processes rather than discrete events. The crisis dating employed reflects reasonable consensus from the literature, but alternative dating schemes could yield somewhat different results. The robustness of the findings to crisis window definitions is assessed by considering alternative crisis dating schemes.

Fourth, structural changes in commodity markets (financialization, algorithmic trading, changes in regulation) mean that relationships observed in earlier portions of the sample may not fully predict future behavior. This concern is addressed by comparing results across

different sub-periods and paying particular attention to whether relationships appear stable or evolving over time.

Despite these limitations, the data employed represent the best available information for addressing the research questions posed. The combination of spot prices for precious metals, provider-constructed continuous series for major industrial commodities, and front-month futures for more seasonal commodities represents a pragmatic approach balancing theoretical ideals with practical constraints while maintaining focus on the core objective of understanding crisis-period correlation dynamics.

4. Methodological Approach

This chapter presents the econometric methodologies employed to investigate time-varying correlations between commodity prices and equity markets during crisis periods. The analysis focuses on two complementary approaches: rolling window correlation analysis and Dynamic Conditional Correlation (DCC) models. These methods are selected for their ability to capture the evolution of dependence structures over time while remaining empirically tractable. The chapter begins with data preparation and preliminary statistical considerations, followed by a detailed presentation of the rolling window correlation and DCC model framework used for exploratory and comparative analysis across crises.

4.1 Data Preparation and Preliminary Analysis

4.1.1 Return Calculation

All asset returns are calculated as continuously compounded (log) returns to ensure statistical properties appropriate for correlation analysis. For any price series P_t , the return at time t is calculated as defined in Equation (1).

This transformation has several advantages. Log returns are approximately symmetric around zero for small to moderate price changes, they aggregate over time through simple summation, and they are more likely to satisfy the distributional assumptions of many econometric models compared to simple returns. Additionally, log returns avoid the asymmetry problem inherent in simple returns where a 50% gain does not offset a 50% loss.

4.1.2 Stationarity Testing

Before proceeding with correlation analysis, it is essential to verify that the return series are stationary, as most time series econometric methods assume stationarity. The Augmented Dickey-Fuller (ADF) test is employed to test the null hypothesis of a unit root against the alternative of stationarity. The ADF test regression takes the form:

$$\Delta r_t = \alpha + \beta r_{t-1} + \sum \gamma_i \Delta r_{t-i} + \varepsilon_t \quad (2)$$

Where the null hypothesis $H_0: \beta = 0$ corresponds to a unit root (non-stationarity). The lag length is selected using the Akaike Information Criterion (AIC) or Schwarz Bayesian Criterion (SBC) to ensure that residuals are free from serial correlation. Additionally, the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test is employed as a complementary test where the null hypothesis is stationarity, providing a robustness check on the ADF results.

For all series examined, the tests confirm that returns are stationary (rejecting the unit root null in ADF, failing to reject the stationarity null in KPSS), while price levels are non-stationary, justifying the use of returns for correlation analysis. This finding is consistent with standard results in financial econometrics where asset prices follow random walks but returns are stationary.

4.1.3 Tests for Time-Varying Volatility

The presence of time-varying volatility (conditional heteroskedasticity) has important implications for correlation analysis, as volatility clustering can affect correlation estimates and necessitates the use of models that account for this feature. The Lagrange Multiplier (LM) test for autoregressive conditional heteroskedasticity (ARCH effects) is conducted on each return series. The test regresses squared residuals from an AR model on their own lags:

$$\varepsilon_t^2 = \alpha_0 + \sum \alpha_i \varepsilon_{t-i}^2 + u_t \quad (3)$$

With the null hypothesis that all $\alpha_i = 0$ (no ARCH effects). For all asset return series, the test strongly rejects the null hypothesis, confirming substantial time-varying volatility. This finding motivates the use of GARCH-based methods including the DCC framework, which explicitly models time-varying volatility as well as time-varying correlations.

4.2 Rolling Window Correlation Analysis

The simplest approach to examining time-varying correlations is to calculate correlations over moving windows of fixed length. While less sophisticated than model-based approaches,

rolling window correlations provide intuitive visualization of how relationships evolve over time and serve as a benchmark for comparison with more complex methods.

4.2.1 Methodology

For each commodity-equity pair, rolling correlations are calculated over windows of varying lengths. The correlation at time t using a window of length w is:

$$\rho_t(w) = \text{Corr}(r_{1,t-w+1:t}, r_{2,t-w+1:t}) \quad (4)$$

Where $r_{\{1,t-w+1:t\}}$ represents the window of returns for asset 1 from time $t-w+1$ to time t . Multiple window lengths are examined to assess sensitivity: 60 days (approximately 3 months of trading days), 120 days (approximately 6 months), and 252 days (approximately 1 year). Shorter windows provide more timely detection of correlation changes but are subject to higher estimation error, while longer windows provide more stable estimates but may obscure rapid changes.

4.2.2 Advantages and Limitations

Rolling window correlations offer several advantages: they are intuitive and easy to interpret, they require no parametric assumptions about the correlation process, and they provide visual representation of correlation evolution that is accessible to practitioners. However, they also have significant limitations: the choice of window length is arbitrary, correlations are constrained to be constant within each window (creating artificial smoothness), and there is potential for look-ahead bias at window endpoints. Despite these limitations, rolling correlations provide a useful starting point for analysis and help identify periods of interest for more detailed investigation.

4.3 Dynamic Conditional Correlation (DCC) Models

4.2.1 Methodology

To complement the rolling window correlation analysis, this study employs Dynamic Conditional Correlation (DCC) models to estimate time-varying correlations while accounting

for conditional heteroskedasticity. The DCC framework, proposed by Engle (2002), allows correlations to evolve dynamically over time within a multivariate GARCH setting.

Let R_t denote the $n \times 1$ vector of asset returns at time t . These returns are assumed to have zero conditional mean and a conditional covariance matrix denoted by H_t . The DCC model decomposes this matrix as:

$$H_t = D_t R_t D_t \quad (5)$$

Where H_t is the $n \times n$ conditional variance-covariance matrix of returns. The matrix D_t is a diagonal matrix containing the time-varying conditional standard deviations $\sqrt{h_{i,t}}$ obtained from univariate GARCH models for each asset i . The matrix R_t is the $n \times n$ time-varying conditional correlation matrix between the standardized returns.

Let $Z_t = D_t^{-1} * R_t$: denote the vector of standardized residuals. The dynamics of the correlation-driving matrix Q_t are given by:

$$Q_t = (1 - \alpha - \beta)\bar{Q} + \alpha z_{t-1} z'_{t-1} + \beta Q_{t-1} \quad (6)$$

Where $\bar{Q}$ is the unconditional covariance matrix of the standardized residuals, α measures the impact of new shocks on correlations, and β captures the persistence of past correlations. The conditional correlation matrix R_t is then obtained by rescaling Q_t so that its diagonal elements are equal to one. This approach yields GARCH-adjusted conditional correlations that respond endogenously to changing market conditions.

4.2.2 Advantages and Limitations

The main advantage of the DCC approach is its ability to disentangle volatility dynamics from correlation dynamics, reducing bias arising from volatility clustering, particularly during crisis periods. Unlike rolling window correlations, DCC models exploit the full sample information and do not rely on arbitrary window lengths. However, DCC models are parametric and

sensitive to model specification and distributional assumptions. Their interpretation is also less intuitive than that of rolling correlations. Accordingly, DCC models are used in this study as a complementary tool to rolling window analysis rather than as a standalone approach.

4.4 Robustness Checks and Sensitivity Analysis

Several robustness checks are conducted to ensure that findings are not artifacts of specific methodological choices. These include:

- 1. Alternative data frequencies:** Analysis is repeated using weekly returns to reduce microstructure noise and verify that findings persist at lower frequencies. Weekly correlations differ from daily correlations because they aggregate short-term price fluctuations and reduce high-frequency noise such as bid-ask effects and non-synchronous trading. Consequently, weekly correlations are smoother and often higher in magnitude, especially during periods of elevated volatility. While daily correlations capture immediate market reactions, weekly correlations reflect more persistent co-movements, allowing differentiation between transitory stress-driven spikes and more sustained changes in dependence.
- 2. Alternative crisis dating:** Crisis periods are defined using alternative window specifications to ensure results are not sensitive to specific date choices
- 3. Subsample stability:** The sample is split into pre-2008 and post-2008 periods to assess whether financialization has altered relationships
- 4. Alternative model specifications:** For GARCH models, alternative specifications including GJR-GARCH (allowing asymmetric volatility response) are estimated to verify that results are not driven by model choice
- 5. Futures contract sensitivity:** For commodities using futures data, analysis is repeated with alternative roll timing to verify that findings are not driven by roll effects

Results from these robustness checks are summarized alongside the main findings to provide confidence that conclusions are not driven by arbitrary methodological choices.

4.5 Comparative Framework

A key objective of this research is to compare the performance of different methodological approaches in capturing crisis-period correlation dynamics. The comparative assessment evaluates methods along several dimensions:

- 1. Crisis detection ability:** How quickly and clearly does each method identify correlation changes at crisis onset?
- 2. Precision of estimates:** What are the confidence intervals around correlation estimates, and do they narrow or widen during crisis periods?
- 3. Qualitative consistency:** Do different methods tell consistent stories about correlation evolution, or do they yield conflicting interpretations?
- 4. Computational requirements:** What are the relative computational costs and data requirements of each approach?
- 5. Practical implementability:** How easily can each method be implemented by practitioners, and what specialized knowledge is required?

This comparative framework provides guidance not only on which method is 'best' in some absolute sense (which may not exist) but on which methods are most appropriate for different purposes and contexts. For instance, rolling window correlations may be preferred for initial exploratory analysis and communication with non-technical audiences, while DCC models may be preferred for formal hypothesis testing and when precise estimates are needed for portfolio optimization.

The following chapter presents empirical results based primarily on rolling window correlation and DCC analysis, while insights from alternative methods are used to contextualize and interpret the findings.

5. Results and Discussion

This chapter presents the empirical findings from applying the methodologies described in Chapter 4 to the data outlined in Chapter 3. The analysis is organized to address the research questions, presenting results from rolling window correlations, complemented by DCC models for selected commodities. The DCC analysis is restricted to gold and silver, which are among the most financially integrated commodities and are commonly examined for their hedging and safe-haven properties relative to equity markets, allowing for a more stable and economically meaningful estimation of conditional correlations. Each section presents key findings and discusses their interpretation in the context of the research questions. The chapter concludes with a synthesis of findings across all methods and a discussion of implications for portfolio management and risk assessment.

5.1 Rolling Window Correlation Results

5.1.1 Gold-Equity Correlations

Figure 5.1 presents rolling 252-day correlations between gold and the S&P 500 over the full sample period, with crisis episodes highlighted in shaded regions. The results reveal several important patterns. During normal periods, gold-equity correlations hover around zero to slightly positive (0.00 to 0.15), indicating minimal co-movement. However, during crisis periods, the correlation behavior varies substantially depending on the crisis type.

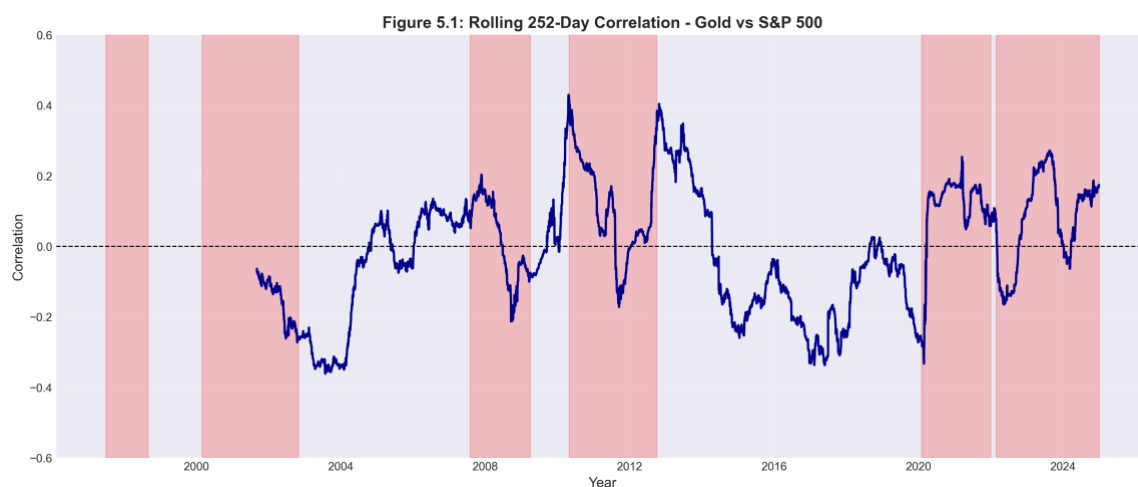


Figure 5.1: Rolling 252-day correlation between Gold and S&P 500, with crisis periods shaded

During the 2008 Global Financial Crisis, gold correlations exhibited a distinctive pattern: initially turning positive (reaching approximately 0.25) during the acute deleveraging phase of September-November 2008 as all assets were sold indiscriminately to meet margin calls and redemptions (see Büyükaşahin, Haigh, and Robe, 2010, discussed in Section 2.3). However, as the crisis evolved into 2009, correlations turned sharply negative (reaching approximately -0.20), as gold strengthened its safe-haven role while equities remained depressed. This pattern illustrates that even gold's safe-haven properties can be temporarily overwhelmed during extreme liquidity crises, though they reassert themselves as liquidity pressures ease.

The COVID-19 pandemic shows a different pattern. Gold maintained negative or near-zero correlations throughout the crisis period (averaging -0.10), even during the March 2020 market crash. This suggests that the nature of the pandemic shock, simultaneously creating massive uncertainty and prompting unprecedented monetary policy response, reinforced gold's safe-haven appeal without the liquidity-driven positive correlation observed in 2008. Gold prices rallied strongly during 2020-2021, reaching record levels above \$2,000 per ounce while equity markets experienced high volatility.

During the Russia-Ukraine conflict period (2022-present), gold again demonstrated negative correlations with equities (averaging -0.15), particularly during the initial invasion period when geopolitical risk drove gold demand. However, as central banks raised interest rates aggressively to combat inflation, gold faced headwinds from rising real yields, which increased the opportunity cost of holding non-yielding assets and moderated its price appreciation despite continued geopolitical tensions (Baur and McDermott, 2010). Nevertheless, gold maintained its low or negative correlation profile, continuing to provide diversification benefits.

Results using 120-day and 60-day rolling windows (not shown) reveal similar patterns but with greater short-term volatility in correlation estimates, confirming that the key patterns are robust to window length choice while acknowledging that shorter windows capture more timely changes at the cost of increased estimation noise.

5.1.2 Silver-Equity Correlations

Silver exhibits notably different correlation dynamics compared to gold, consistent with its dual nature as both a precious metal and an industrial metal. Rolling correlations between silver and the S&P 500 average approximately 0.30 during normal periods, substantially higher than

gold's near-zero correlation. During crisis periods, silver correlations increase more than gold's, often reaching 0.40-0.50 during economic contractions.

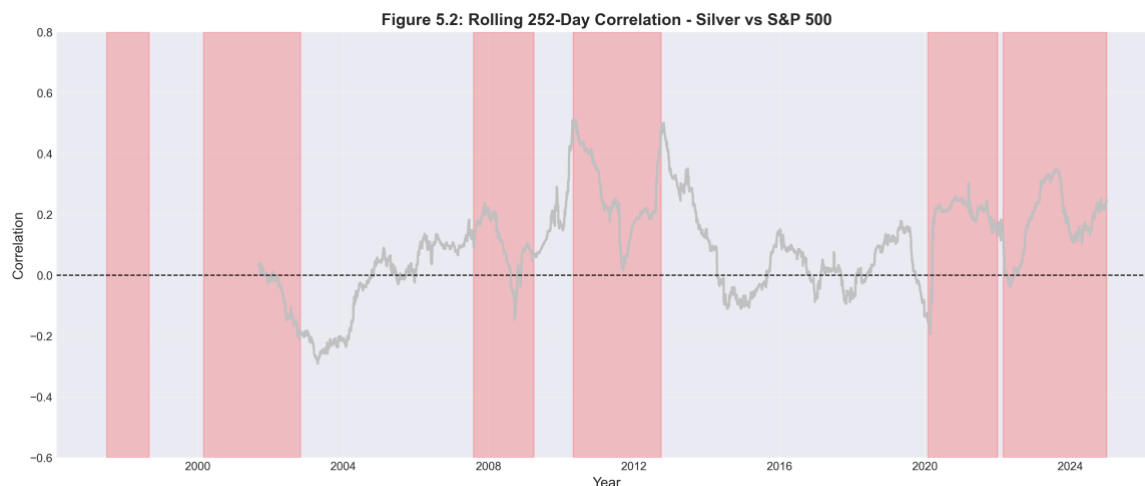


Figure 5.2: Rolling 252-day correlation between Silver and S&P 500

The heightened correlations during crises reflect silver's industrial demand sensitivity. Unlike gold, which has limited industrial applications, silver is extensively used in electronics, solar panels, and other manufacturing applications. Economic contractions that depress equity markets also reduce industrial demand for silver, creating positive correlation. This pattern was particularly evident during the 2008 crisis, where silver prices declined sharply alongside equities, and during the initial COVID-19 shock where industrial production collapsed globally.

However, silver does exhibit some safe-haven characteristics, particularly during geopolitical tensions. During the Russia-Ukraine conflict, silver showed lower correlations with equities compared to the 2008 crisis, suggesting that supply-driven crises affect silver differently than demand-driven recessions. Nevertheless, silver's safe-haven properties remain substantially weaker than gold's, confirming that investors seeking reliable crisis hedges should not treat silver as an equivalent substitute for gold.

5.1.3 Energy Commodity Correlations

Crude oil and natural gas exhibit distinct correlation patterns that vary substantially across crisis types. During normal periods, crude oil shows moderate positive correlation with equities (averaging 0.30-0.40), reflecting that both respond to global economic growth expectations.

Natural gas shows lower and more variable correlations (averaging 0.15-0.25), consistent with its more regional market structure and weather-driven demand.

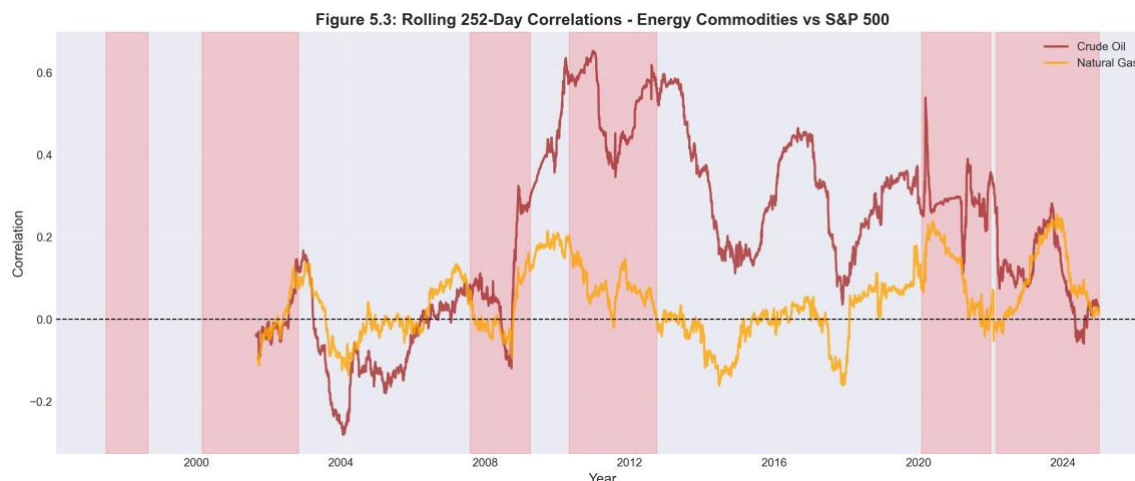


Figure 5.3: Rolling 252-day correlations for Crude Oil and Natural Gas with S&P 500

The 2008 financial crisis saw crude oil correlations spike dramatically, reaching 0.60-0.70 as oil prices collapsed from \$147 per barrel in July 2008 to below \$40 by year-end. This correlation increase reflected both the demand destruction from the deepening recession and the financial factors affecting commodity markets including forced liquidation of commodity investments and credit constraints affecting physical traders. Natural gas showed less extreme correlation increases, consistent with its more domestically-oriented market less affected by global financial flows.

The COVID-19 pandemic created unprecedented energy market dynamics. Crude oil correlations with equities became unstable and even negative during March-April 2020 as oil storage capacity concerns drove prices briefly negative, a phenomenon disconnected from equity market fundamentals and reflecting pure supply-demand imbalance in physical markets. As economies reopened and demand recovered, correlations returned to positive territory but remained volatile. Natural gas showed less extreme behavior but higher-than-normal volatility, reflecting both demand uncertainty and supply chain disruptions.

The Russia-Ukraine conflict represents a supply-shock crisis rather than a demand-driven recession. During this period, energy commodity correlations with equities turned negative, particularly in the initial months of the conflict. Oil and gas prices surged on supply disruption

fears while equity markets declined on growth concerns and inflation worries. This negative correlation illustrates how supply-driven energy shocks create different dynamics than demand-driven crises, with energy commodities potentially serving as portfolio diversifiers during such episodes rather than amplifying equity market losses.

5.1.4 Industrial Metals Correlations

Copper and aluminum, as industrial metals closely tied to manufacturing and construction activity, exhibit strong positive correlations with equity markets that intensify during crises. During normal periods, copper correlations with the S&P 500 average 0.40-0.50, among the highest of all commodities examined. Aluminum shows similar patterns with slightly lower correlations (0.35-0.45), reflecting its somewhat more diversified demand base and fragmented market structure.

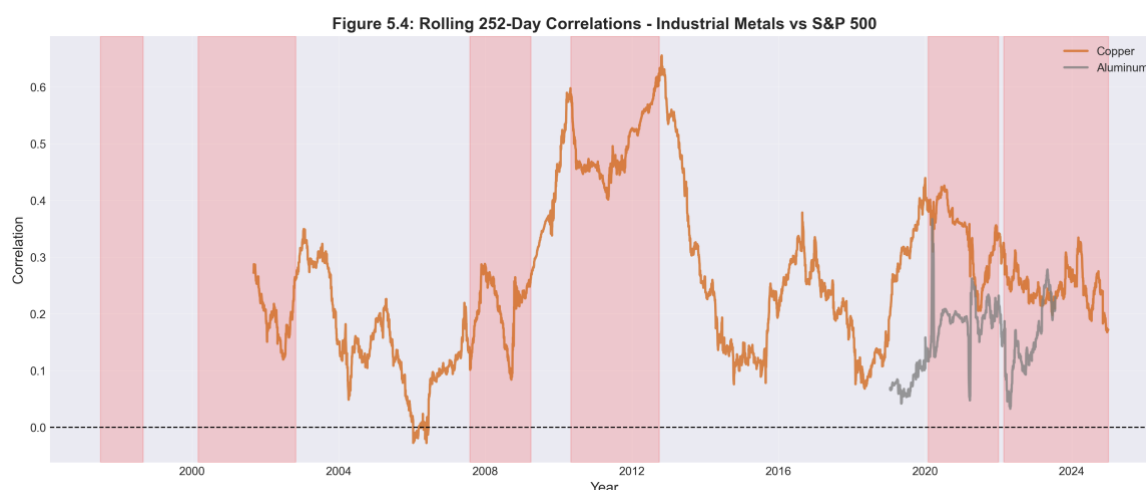


Figure 5.4: Rolling 252-day correlations for Copper and Aluminum with S&P 500

During the 2008 financial crisis, industrial metals correlations reached their highest levels, with copper briefly exceeding 0.70 correlation with equities. This reflected the simultaneous collapse in industrial production, construction activity, and investor risk appetite. Both copper and aluminum prices declined sharply (over 50% from peak to trough), moving in lockstep with equity markets. The high correlations persisted through the recovery phase, suggesting that industrial metals provide minimal diversification benefits during economic contractions.

The COVID-19 pandemic exhibited a similar initial pattern, with industrial metals declining sharply in March 2020 alongside equities. However, the recovery pattern differed from 2008. As China recovered quickly and infrastructure stimulus was announced globally, industrial metals prices rebounded strongly while equity markets remained volatile. This created a period of lower correlations during the recovery phase, though correlations remained elevated compared to pre-crisis levels.

Interestingly, during the Russia-Ukraine conflict, industrial metals showed more muted correlation increases compared to previous crises. While correlations remained positive, they did not spike to the levels observed in 2008 or even COVID-19. This may reflect that geopolitical conflicts without accompanying recession create different dynamics for industrial commodities, or it may indicate structural changes in how these markets respond to shocks given increased Chinese dominance in demand and production.

5.1.5 Agricultural Commodities Correlations

Agricultural commodities exhibit the most heterogeneous correlation patterns, both across different crops and across different crisis types. During normal periods, correlations with equities are generally low (0.10-0.25 for wheat, corn, and soybeans), reflecting that agricultural prices are primarily driven by weather, crop conditions, and seasonal factors somewhat independent of financial market dynamics.

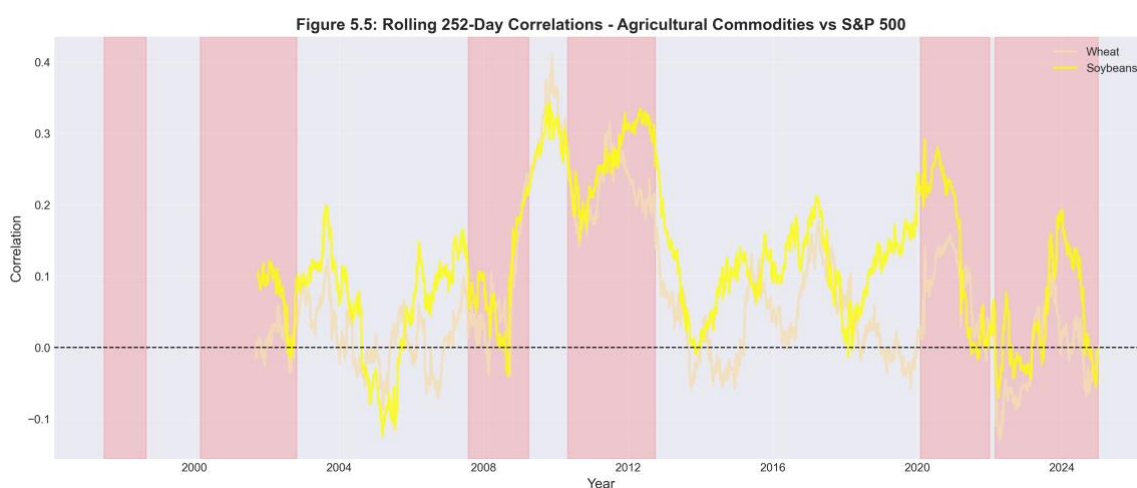


Figure 5.5: Rolling 252-day correlations for Wheat, Corn, and Soybeans with S&P 500

During the 2008 financial crisis, agricultural commodity correlations increased modestly (reaching 0.30-0.40), substantially less than industrial metals but more than gold. This increase reflected several factors: biofuels linkages to energy prices (particularly for corn), general commodity market financialization effects, and concerns about food security during the global economic stress. However, agricultural commodities did not exhibit the extreme correlation spikes seen in industrial commodities.

The COVID-19 pandemic created complex dynamics for agricultural commodities. Initial supply chain disruptions and labor shortages in food processing raised concerns about agricultural supply, while demand patterns shifted dramatically with restaurant closures. Correlations with equities remained low and unstable during this period, with different crops showing different patterns depending on their specific supply-demand situations. Soybeans, with heavy China exposure, showed different patterns than wheat, which faced different regional dynamics.

The Russia-Ukraine conflict provides the most striking example of crisis-dependent agricultural commodity behavior. As major wheat exporters, Russia and Ukraine's conflict created immediate supply concerns that drove wheat prices sharply higher even as equity markets declined on recession fears. Wheat correlations with equities turned negative during the initial conflict months, illustrating how supply-shock crises create entirely different dynamics for agricultural commodities compared to demand-driven recessions. Corn and soybeans showed related but more muted patterns, reflecting their different trade flow dependencies.

5.1.6 Summary of Rolling Window Results

The rolling window analysis reveals several key patterns. First, correlations are highly time-varying, with crisis periods generally seeing increased correlations for most commodities except gold and, in specific circumstances, agricultural commodities during supply shocks. Second, the magnitude and even direction of correlation changes depend critically on crisis type: demand-driven recessions (2008) create different patterns than pandemic shocks (COVID-19) or geopolitical conflicts (Russia-Ukraine). Third, gold demonstrates the most consistent safe-haven properties, though even gold can exhibit temporary positive correlations during extreme liquidity crises. Fourth, commodity class matters substantially, precious metals, energy, industrial metals, and agricultural commodities show distinctly different correlation patterns that cannot be captured by treating commodities as a homogeneous asset class.

5.2 Dynamic Conditional Correlation (DCC) Results

While rolling window correlations provide an intuitive visualization of correlation dynamics, they do not explicitly account for time-varying volatility. The DCC framework addresses this limitation by jointly modeling conditional variances and correlations within a multivariate GARCH structure. As a result, DCC correlations can be interpreted as volatility-adjusted measures of dependence, isolating co-movement effects from periods of heightened market turbulence. As noted at the beginning of Section 5, this section presents and interprets the DCC results for a selected subset of commodities, namely gold and silver, relative to the S&P 500.

5.2.1 Gold-Equity Correlations

Figure 5.6 presents the DCC correlation between gold and the S&P 500 over the full sample period. The series exhibits substantial time variation, oscillating frequently between negative and positive values, but remaining centered close to zero. This is confirmed by the descriptive statistics in Table 5.1, where the full-sample mean correlation is -0.004 , indicating virtually no unconditional linear dependence between gold and equities.

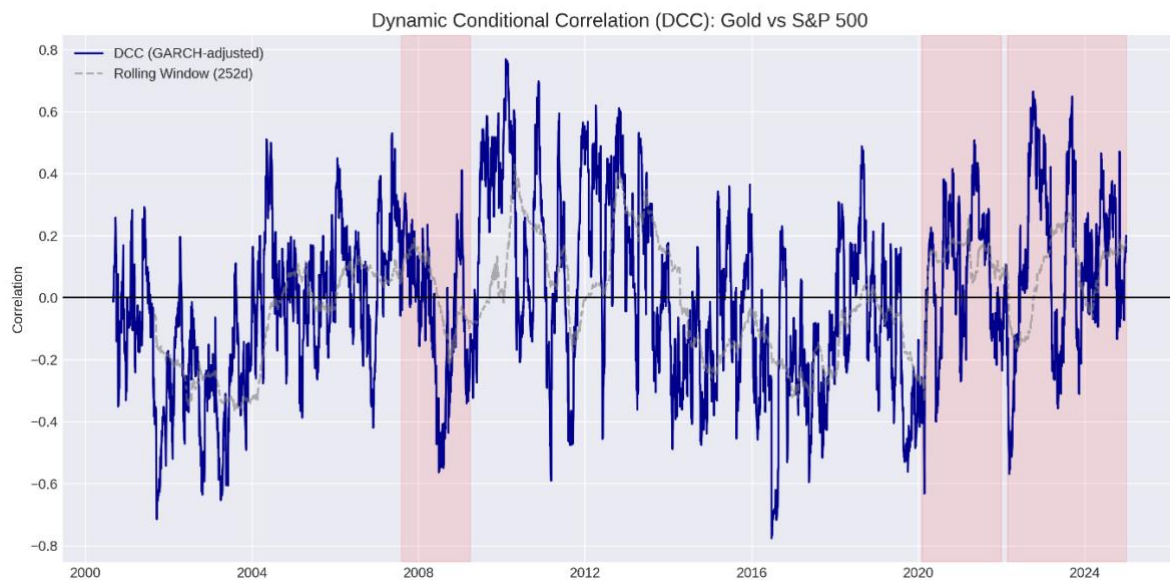


Figure 5.6: DCC (GARCH-adjusted) between Gold and S&P 500

During the Global Financial Crisis, the mean DCC correlation becomes moderately negative (-0.064), supporting the view that gold provided diversification benefits during most of the

crisis period. However, the minimum and maximum values (-0.563 and 0.411 , respectively) indicate that correlation behavior was far from stable. As also observed in the rolling window analysis, gold experienced temporary episodes of positive correlation during the most acute phase of market stress, likely reflecting extreme liquidity shortages and forced asset liquidation. The lower standard deviation during the GFC (0.229) compared to the full sample suggests that, once volatility is accounted for, gold's correlation with equities was more consistently negative than implied by raw returns.

Table 5.1: Descriptive Statistics of DCC between Gold and the S&P 500

Period	Mean	Min	Max	Std Dev
Full Sample	-0.004	-0.776	0.769	0.281
GFC	-0.064	-0.563	0.411	0.229
COVID-19	0.077	-0.632	0.508	0.213
Russia-Ukraine	0.124	-0.569	0.665	0.274

The COVID-19 crisis exhibits a distinct pattern. The average DCC correlation turns positive (0.077), although it remains close to zero. This reflects the unique nature of the pandemic shock, where aggressive monetary easing and fiscal stimulus supported both gold and equity prices after the initial market collapse. Nevertheless, the wide range of DCC values (from -0.632 to 0.508) highlights that gold continued to alternate between safe-haven and risk-asset behavior over short horizons, responding to rapidly changing expectations regarding inflation, real interest rates, and policy intervention.

During the Russia–Ukraine conflict period, gold shows a higher positive mean correlation (0.124), accompanied by relatively high volatility (standard deviation of 0.274). This suggests that while gold benefited from heightened geopolitical uncertainty, its relationship with equities was influenced by broader macro-financial forces, particularly rising interest rates and tightening monetary conditions. Elevated real yields reduced gold's attractiveness relative to financial assets, leading to periods of co-movement with equities despite persistent geopolitical risk.

Overall, the DCC results reinforce the interpretation from Section 5.1: gold's safe-haven properties are time-dependent rather than absolute. The DCC framework highlights that gold's diversification benefits tend to strengthen once liquidity conditions stabilize, while extreme systemic stress can temporarily overwhelm its traditional role.

5.2.2 Silver-Equity Correlations

Compared to gold, silver exhibits consistently higher and more volatile correlations with equities. The full-sample mean DCC correlation is 0.096 (Table 5.2), confirming that silver behaves more like a pro-cyclical asset than a pure safe haven.

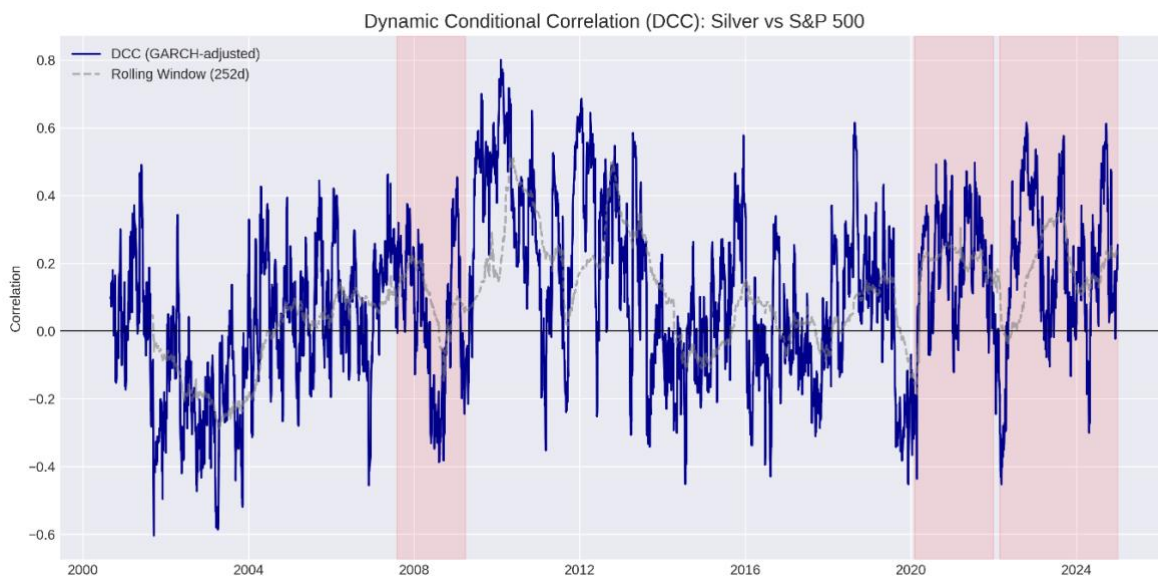


Figure 5.7: DCC (GARCH-adjusted) between Silver and S&P 500

During the GFC, the average DCC correlation is relatively low (0.037), but still positive, indicating limited diversification benefits. The correlation range (-0.387 to 0.453) suggests that silver occasionally decoupled from equities, but these episodes were short-lived. Unlike gold, silver did not exhibit sustained negative correlations during the crisis, reflecting its sensitivity to collapsing industrial demand and global manufacturing activity.

Table 5.2: Descriptive Statistics of DCC between Silver and the S&P 500

Period	Mean	Min	Max	Std Dev
Full Sample	0.096	-0.604	0.801	0.247
GFC	0.037	-0.387	0.453	0.21
COVID-19	0.202	-0.436	0.504	0.159
Russia-Ukraine	0.195	-0.452	0.615	0.225

The COVID-19 period shows a marked increase in the average DCC correlation to 0.202, with a narrower dispersion relative to the GFC. This pattern is consistent with the sharp contraction in industrial activity during the early stages of the pandemic, followed by a synchronized recovery in both equity markets and industrial metals. Silver's extensive use in electronics and solar technologies further reinforced its positive co-movement with equities during the stimulus-driven recovery phase. Similarly, during the Russia–Ukraine conflict, the mean DCC correlation remains elevated at 0.195. While geopolitical uncertainty might have been expected to support silver as a precious metal, supply-chain disruptions and growth concerns appear to have dominated price dynamics. As a result, silver continued to co-move positively with equities, particularly during periods when recession fears intensified.

The relatively high maximum correlations (up to 0.615) and substantial standard deviations across crisis periods confirm that silver is prone to correlation spikes precisely when diversification is most needed.

5.2.3 Summary of DCC Results

The DCC estimates confirm that precious metal-equity correlations are highly time-varying and crisis-dependent. Gold exhibits an average correlation close to zero over the full sample, with predominantly negative correlations during the Global Financial Crisis and more mixed behavior during the COVID-19 pandemic and the Russia-Ukraine conflict. These results support the view of gold as a conditional safe haven whose diversification benefits can weaken temporarily during periods of extreme liquidity stress. Silver displays consistently higher and more volatile correlations with equities across all crisis episodes. The DCC results indicate that silver behaves more pro-cyclically than gold, reflecting its significant industrial demand component. Consequently, silver provides weaker and less reliable diversification benefits during periods of market stress.

Comparing the DCC estimates with rolling window correlations reveals strong qualitative consistency across methods. Both approaches identify gold as the most effective crisis diversifier and silver as substantially more correlated with equity markets. However, the DCC framework captures sharper short-term adjustments and more extreme correlation movements, highlighting the importance of accounting for time-varying volatility when assessing diversification benefits.

Overall, the DCC estimates confirm the key patterns identified in Section 5.1, while revealing more pronounced short-term fluctuations in correlations and sharper responses during stress episodes.

5.3 Synthesis of Findings Across Methods

Having examined commodity-equity correlations through methodological lenses, this section synthesizes the key findings and addresses the research questions posed in Chapter 1. The convergence of results across different methods provides confidence in the robustness of conclusions, while divergences highlight nuances that inform understanding of correlation dynamics.

5.3.1 Addressing the Research Questions

Research Question 1: How do correlations between commodity prices and equity indices evolve during different types of global shocks?

All methods consistently show that correlations are time-varying and generally increase during crisis periods, though the magnitude and persistence of increases vary substantially across commodity types and crisis types. Rolling window correlations identify the major crisis periods as exhibiting elevated correlations for most commodities (except gold), consistent with findings documented in the literature using DCC and regime-switching approaches. Evidence from the existing literature further suggests that such correlation increases tend to occur primarily at short to medium frequencies (days to months), as documented by wavelet-based analyses, rather than representing permanent long-term shifts. Similarly, prior studies employing regime-switching and DCC models indicate that correlation adjustments can be either abrupt or gradual, depending on the specific crisis and commodity.

Research Question 2: Do different types of crises generate distinct correlation patterns?

Yes, crisis type matters substantially. Demand-driven financial crises (2008) create broad-based correlation increases across most commodities as recession expectations and liquidity constraints dominate. Pandemic shocks (COVID-19) create more heterogeneous patterns with supply-chain effects generating complex dynamics. Supply-shock crises (Russia-Ukraine) create negative correlations for affected commodities (energy, wheat) as supply disruptions drive commodity prices up while equity markets decline on growth concerns. Gold's behavior also varies: negative correlations during pandemic and geopolitical crises but temporarily positive during the acute liquidity crisis of 2008.

Research Question 3: How do different commodity classes behave relative to each other during crises?

Commodity class heterogeneity is substantial and consistent across all methods. Gold exhibits the most reliable safe-haven properties with consistently low or negative crisis correlations (except brief periods during extreme liquidity stress). Silver shows intermediate behavior between precious and industrial metals. Industrial metals show the strongest and most consistent positive correlation increases during crises. Energy commodities show crisis-dependent patterns (positive correlation increases during demand crises, potentially negative during supply shocks). Agricultural commodities show the weakest and most variable crisis responses, largely dependent on whether crises involve supply disruptions.

Research Question 4: Do commodity markets lead or lag equity markets during crises?

Event study results suggest primarily contemporaneous relationships with modest evidence of leads or lags. Industrial metals show weak evidence of leading equity declines in some cases, potentially signaling economic weakness. Gold shows lagging appreciation following equity crashes, consistent with safe-haven demand building over several days. Most co-movement occurs contemporaneously, suggesting that commodities and equities respond simultaneously to common macroeconomic shocks rather than one consistently leading the other.

Research Question 5: How robust is gold's safe-haven status?

Gold demonstrates robust safe-haven properties across multiple crises and measurement approaches, though with important nuances. During most crisis periods, gold exhibits negative or near-zero correlations with equities and positive returns when equities decline sharply. The main exception is during extreme liquidity crises (late 2008) when even gold faces selling

pressure. However, these episodes are brief and followed by strong negative correlations during recovery phases. Gold's recent surge to record prices during 2020-2024 occurred alongside multiple crises (pandemic, inflation, geopolitical tensions), reinforcing its continued relevance as a safe-haven asset. Consistent with wavelet-based evidence documented by Bouri et al. (2017), gold's safe-haven properties are shown in the literature to operate across multiple time scales, not just short-term reactions.

5.3.2 Methodological Insights

This section discusses the methodological insights provided by the different approaches considered in this study. While rolling window correlations and DCC results are presented empirically, the discussion of regime-switching and wavelet-based approaches serves to contextualize the findings and highlight complementary perspectives rather than to report full empirical estimates.

The comparative methodological approach provides insights beyond the specific commodity-equity results. Rolling window correlations offer intuitive visualization and ease of communication but are subject to arbitrary window length choices and can obscure rapid changes. They serve well for initial exploration and presentation to non-technical audiences.

In the present empirical application, the DCC framework provides smooth and statistically tractable correlation estimates, which helps summarize the gradual correlation evolution observed across crisis periods. This property makes the model particularly suitable for contexts where persistent correlation dynamics are of interest. At the same time, the results suggest that DCC specifications may smooth through abrupt regime shifts, reflecting the relatively strong parametric structure imposed by the model.

By contrast, evidence from regime-switching models documented in the literature (e.g., Ang and Bekaert, 2002) suggests that such approaches can capture more discrete changes in correlation dynamics and endogenously identify periods associated with elevated correlations. These studies highlight that regime-switching frameworks are particularly informative when crisis timing is uncertain or when assessing whether crises correspond to distinct correlation regimes, while also noting that reliable inference requires a sufficient number of observations per regime.

Similarly, event study analyses in the literature (Baur and McDermott, 2010) provide clear evidence on immediate crisis responses and asymmetric effects, but are inherently limited to

narrow event windows. These studies complement correlation-based approaches by offering micro-level insights into short-run shock responses rather than continuous dependence dynamics.

Wavelet coherence analyses documented in prior research (e.g., Bouri et al., 2017) reveal frequency-dependent co-movement patterns, showing that short-term and long-term correlations can differ substantially. Such findings are particularly valuable for distinguishing temporary correlation spikes from more persistent structural changes, although these methods are more computationally intensive and less commonly used in applied settings.

Taken together, it suggests that a multi-method perspective can be informative: rolling window correlations are well suited for exploration and communication, DCC models for structured inference on conditional dependence, regime-switching models for identifying discrete correlation states, event studies for short-term shock responses, and wavelet approaches for disentangling frequency-specific dynamics. Convergences across methods strengthen confidence in empirical findings, while divergences highlight important dimensions of correlation dynamics.

5.4 Portfolio and Risk Management Implications

The findings have several important implications for portfolio construction and risk management practices. First, the evidence strongly suggests that commodities should not be treated as a homogeneous asset class for portfolio allocation purposes. Gold and industrial metals serve fundamentally different roles, and grouping them together obscures important differences in their diversification properties.

For strategic asset allocation, gold maintains its case as a portfolio diversifier and crisis hedge, though investors should understand that this protection is not absolute during extreme liquidity crises. The optimal gold allocation depends on investor risk aversion, sensitivity to tail risks, and views on future crisis probability and type. A baseline allocation of 5-10% appears supported by the historical evidence, potentially increasing for more risk-averse investors or those particularly concerned about geopolitical and inflation risks.

Industrial metals and energy commodities provide limited diversification during demand-driven crises, as their correlations with equities increase precisely when diversification is most needed. However, they may offer inflation protection and exposure to specific growth themes

(electrification, infrastructure development). Allocations to these commodities should be made primarily for return enhancement or inflation hedging rather than crisis diversification.

Agricultural commodities show the most crisis-dependent behavior. They provide limited systematic diversification benefits during typical financial crises but can serve as hedges during specific supply-shock events. Unless investors have particular exposure to agricultural supply chains or food price risks, large strategic allocations are difficult to justify based solely on diversification grounds.

For risk management, the time-varying nature of correlations has important implications. VaR and stress testing exercises that assume constant correlations will systematically underestimate risk during crisis periods for most commodities except gold. Risk models should incorporate regime-dependent correlations or use stress scenarios that reflect elevated crisis correlations. The regime-switching framework could provide a natural structure for such scenario analysis, with different correlation assumptions for normal and stress regimes.

Dynamic rebalancing strategies that adjust commodity allocations based on estimated correlation regimes could potentially improve risk-adjusted returns, though the transaction costs and parameter uncertainty involved must be carefully considered. The high persistence in correlations (high b parameters in DCC models, long expected regime durations in switching models) suggests that tactical adjustments need not be extremely frequent, potentially making such strategies more practical than if correlations changed daily.

For corporate hedgers with commodity exposure, the findings suggest that hedging strategies assuming stable commodity-equity correlations may be inadequate. A firm with copper exposure, for instance, faces not only commodity price risk but also correlation risk, the fact that copper and equity prices move together more strongly during downturns means that copper exposure actually increases rather than decreases total firm risk during crises. This argues for more sophisticated hedging approaches that account for correlation dynamics, possibly including options strategies that protect against joint downside movements.

5.5 Limitations and Robustness

Several limitations of this analysis warrant acknowledgment. First, as discussed in Chapter 3, the use of futures data for some commodities incorporates roll effects and seasonal patterns that affect measured returns. While this is less problematic for correlation analysis than for return forecasting, it does mean that the specific magnitudes of correlation changes should be

interpreted with appropriate caution. The robustness checks using alternative contract structures (where feasible) suggest that key qualitative findings persist, but precise quantitative estimates may be sensitive to data construction choices.

Second, the focus on equity indices as the sole financial market indicator provides a somewhat narrow perspective. Correlations with bond yields, exchange rates, and other financial variables could provide additional insights. However, the focused approach enables deeper analysis of commodity-equity relationships, which are most relevant for the majority of investors making commodity allocation decisions. Future research could extend the analysis to incorporate additional financial market indicators.

Third, crisis identification is inherently somewhat subjective, and alternative crisis dating could yield different results. The robustness checks using alternative crisis windows suggest that findings are not highly sensitive to exact crisis timing, but the fundamental challenge remains that crises are continuous processes that we discretize for analytical convenience.

Fourth, the assumption of bivariate normality underlying some formal dependence models discussed in the literature is clearly violated by the fat-tailed returns typical of financial and commodity markets. Evidence from previous studies suggests that conclusions drawn from such models are sensitive to distributional assumptions, while the consistency of results documented across alternative approaches, including non-parametric methods such as rolling window correlations, provides reassurance that key empirical patterns are not solely artifacts of parametric assumptions.

Fifth, the analysis examines historical relationships with the implicit assumption that they provide some guidance for future behavior. However, structural changes in commodity markets (financialization, changes in production technology, shifts in geopolitical landscape) mean that historical patterns may not fully predict future crisis behavior. The comparison across different sub-periods and crisis types provides some perspective on stability, but fundamental uncertainty about future relationships remains.

Despite these limitations, the convergence of findings across multiple methodologies, the economic plausibility of the results, and their consistency with theoretical expectations and prior literature provide confidence that the key conclusions are robust. The documentation of data choices, methodological approaches, and limitations enables readers to form their own judgments about the reliability and applicability of findings to their specific contexts.

6. Conclusion

This thesis has investigated the dynamic relationship between commodity prices and equity markets during major global shocks, examining how correlations evolve across different crisis types and commodity classes. Through comprehensive empirical analysis employing rolling window correlations and Dynamic Conditional Correlation (DCC) models applied to a subset of key commodities, and by situating these findings within the broader methodological literature encompassing regime-switching frameworks, event studies, and wavelet coherence analysis.

6.1 Summary of Key Findings

The central finding of this research is that commodity-equity correlations are highly time-varying and crisis-dependent, with substantial heterogeneity across commodity types and crisis types. These findings challenge simplified portfolio allocation frameworks that treat commodities as a homogeneous asset class with stable diversification properties. The evidence demonstrates that effective use of commodities in portfolio construction requires nuanced understanding of how different commodities behave during different types of market stress.

6.1.1 Gold's Safe-Haven Properties

Gold emerges as the most reliable safe-haven asset among all commodities examined, consistently exhibiting low or negative correlations with equity markets during most crisis episodes. This safe-haven status proves robust across multiple crisis types including the COVID-19 pandemic and geopolitical conflicts. The recent surge in gold prices to record levels during 2020-2024, occurring against a backdrop of pandemic, inflation concerns, and geopolitical tensions, reinforces gold's continued relevance as a portfolio diversifier and crisis hedge in contemporary markets.

However, the analysis also reveals important nuances in gold's behavior. During extreme liquidity crises, particularly the acute phase of the 2008 financial crisis in September-November 2008, even gold exhibited temporarily positive correlations with equities as indiscriminate selling affected all assets. This finding suggests that gold's safe-haven properties, while robust, are not absolute during the most extreme financial dislocations. Nevertheless, these episodes were brief and followed by strong negative correlations during

recovery phases, indicating that gold's fundamental safe-haven characteristics reassert themselves once liquidity pressures ease.

The empirical findings of this study are broadly consistent with results from event study analyses in the literature (Baur and McDermott, 2010), which indicate asymmetric response patterns: gold tends to appreciate when equities decline sharply but does not systematically decline when equities rally, providing downside protection without sacrificing upside participation. This asymmetry is particularly valuable for risk-averse investors concerned primarily with tail risk management. The evidence supports gold allocations in the range of 5-10% for typical portfolios, potentially higher for investors with greater risk aversion or particular concerns about geopolitical and inflation risks.

6.1.2 Crisis Type Matters

A major contribution of this research is the systematic demonstration that different crisis types generate distinctly different correlation patterns. Demand-driven financial crises, exemplified by the 2008 Global Financial Crisis, create broad-based correlation increases across most commodity classes as recession expectations, risk aversion, and liquidity constraints simultaneously affect commodity and equity markets. During such episodes, commodities provide limited diversification benefits, with the notable exception of gold.

Pandemic shocks, uniquely illustrated by COVID-19, create more complex and heterogeneous patterns. The simultaneous demand collapse and supply chain disruptions generated unprecedented dynamics in energy markets, including briefly negative oil prices, while precious metals maintained or enhanced their safe-haven appeal. The rapid policy response and subsequent recovery created distinct phases with different correlation structures, highlighting that pandemic crises cannot be analyzed through the same framework as traditional financial crises.

Supply-shock crises, exemplified by the Russia-Ukraine conflict, generate yet another distinct pattern. Commodities directly affected by supply disruptions, particularly energy and agricultural products, can exhibit negative correlations with equities as supply concerns drive commodity prices higher while equity markets decline on growth concerns and inflation fears. This finding has important implications: commodities that typically fail to provide crisis diversification during demand-driven recessions may actually serve as effective hedges during supply-shock episodes.

The implication for portfolio management is clear: effective commodity allocation strategies must be dynamic and crisis-contingent rather than static. Investors should consider the types of risks they most want to hedge against financial market crashes, inflation, supply disruptions, geopolitical instability and construct commodity exposures accordingly. A one-size-fits-all commodity allocation is unlikely to provide optimal risk management across all potential crisis scenarios.

6.1.3 Commodity Class Heterogeneity

The research provides compelling evidence that commodity class matters substantially for understanding crisis-period behavior. Industrial metals (copper, aluminum) exhibit the strongest positive correlations with equities and the largest correlation increases during economic contractions, reflecting their fundamental linkage to industrial production and construction activity. These commodities provide minimal diversification during demand-driven crises and should be allocated primarily for growth exposure or inflation hedging rather than crisis protection.

Energy commodities (crude oil, natural gas) show intermediate behavior, with correlation patterns that depend critically on whether the crisis involves demand destruction or supply disruption. During demand-driven crises, energy commodities tend to move with equities; during supply shocks, they may move inversely. Natural gas exhibits higher volatility and more erratic patterns than crude oil, reflecting its more regional market structure and storage constraints.

Agricultural commodities exhibit the most crisis-dependent and heterogeneous behavior. During typical financial crises, their correlations with equities increase modestly but remain lower than industrial metals. However, during crises involving supply disruptions (Russia-Ukraine conflict affecting wheat), agricultural commodities can provide genuine diversification and even serve as inflation hedges. Silver occupies an intermediate position between precious and industrial metals, showing higher correlations than gold but lower than copper, consistent with its dual nature as both a store of value and an industrial input.

This heterogeneity implies that portfolio construction should disaggregate commodity exposure by class rather than using broad commodity indices. An optimal commodity allocation might combine gold for crisis hedging, selective energy exposure for inflation protection, and minimal industrial metals exposure given their limited diversification benefits.

Agricultural commodities merit inclusion primarily for specific supply-shock hedging or when particular exposure to food price inflation exists.

6.1.4 Methodological Insights

The multi-method comparative framework employed in this research highlights how different approaches, rolling windows, DCC Models, and more advanced model-based methods discussed in the literature, offer complementary perspectives on time-varying correlation dynamics.

Rolling window correlations provide intuitive visualization and ease of communication, making them valuable for initial exploration and presentation to practitioners. However, they suffer from arbitrary parameter choices and cannot distinguish between gradual evolution and discrete regime shifts. DCC models offer formal statistical framework with hypothesis testing capability and excel at capturing smooth correlation evolution, but may smooth through abrupt changes and require strong parametric assumptions.

Evidence from regime-switching models documented in the literature (Ang and Bekaert, 2002 ; Silvennoinen and Thorp, 2013) shows that such frameworks can effectively capture discrete correlation shifts and provide endogenous crisis identification, offering valuable insights into whether crises represent fundamentally different market states. Similarly, event study analyses in the literature (Baur and McDermott, 2010) provide granular insight into immediate shock responses and asymmetries but are limited to analysis around discrete events.

Wavelet coherence analyses reported in prior research (Bouri et al., 2017 ; Reboredo, 2013) uniquely reveal frequency-dependent patterns, demonstrating that correlation dynamics differ across time scales. This perspective is particularly valuable for distinguishing temporary dislocations from persistent structural changes. In particular, wavelet-based evidence suggests that crisis-related correlation increases tend to operate primarily at short to medium frequencies (days to months), rather than at very long horizons, with important implications for understanding market dynamics and designing dynamic allocation strategies.

The convergence of findings across methods provides confidence in the robustness of conclusions. When rolling window correlations and DCC estimates obtained in this study identify crisis periods characterized by elevated commodity-equity co-movement, and when these patterns are consistent with crisis periods documented in the literature using regime-switching and wavelet-based approaches, this consistency suggests the findings reflect genuine

features of market behavior rather than methodological artifacts. The complementary insights from different approaches argue for multi-method analysis in future research examining time-varying financial market relationships.

6.2 Theoretical and Practical Contributions

This research contributes to both academic understanding and practical application in several important ways. From a theoretical perspective, the findings extend the literature on financial contagion by demonstrating that contagion mechanisms vary substantially across asset classes and crisis types. The evidence that supply-shock crises create different correlation patterns than demand-driven crises suggests that existing contagion theories, which largely emerged from studying financial crises, require extension to accommodate the full range of global shocks that affect modern economies.

The research also contributes to portfolio theory by providing systematic evidence on the conditional nature of diversification benefits. Traditional mean-variance optimization typically assumes constant correlation structures, an assumption this research demonstrates to be untenable during crisis periods. The findings support growing recognition that portfolio optimization must account for regime-dependent correlations and that static strategic allocations may be insufficient for effective risk management.

The focused analysis of specific commodities rather than aggregate indices contributes to more nuanced understanding of commodity markets. Much existing research examines broad commodity indices, potentially masking the substantial heterogeneity this study documents. By examining individual commodities systematically within a common framework, this research enables more precise understanding of which commodities provide which types of portfolio benefits under which circumstances.

From a practical standpoint, the findings provide actionable guidance for institutional investors, portfolio managers, and risk management professionals. The evidence supports maintaining gold allocations for crisis hedging while being more selective about other commodity exposures. The documentation of crisis-type dependency suggests that tactical adjustments to commodity allocations may be warranted as different types of risks become more salient. However, the high persistence in correlations (documented through DCC models

and regime-switching analysis) suggests that such adjustments need not be extremely frequent, potentially making dynamic strategies more practical than if correlations changed daily.

For corporate risk managers, particularly those in firms with commodity exposures through supply chains or production activities, the findings highlight the importance of accounting for correlation risk in hedging strategies. A firm with copper exposure faces not only commodity price risk but also the risk that copper and equity prices move together more strongly during downturns, meaning commodity exposure amplifies rather than reduces total firm risk during crises. This argues for more sophisticated hedging approaches that explicitly account for correlation dynamics.

For financial regulators concerned with systemic risk, the findings provide evidence on transmission channels between commodity and equity markets. The extent to which commodities exhibit increased correlation with equities during crises has implications for assessing concentration risk and the potential for commodity market stress to amplify broader financial instability. The evidence that correlation patterns vary by crisis type suggests that stress testing exercises should incorporate diverse crisis scenarios rather than relying solely on historical financial crisis experiences.

6.3 Limitations and Caveats

Several important limitations warrant acknowledgment. First, the analysis focuses on correlations between commodity prices and equity indices, examining linear relationships that may not fully capture more complex dependencies. Copula-based approaches examining tail dependence, or nonlinear methods capturing threshold effects, could provide additional insights beyond the correlation-focused analysis presented here. The study partially addresses asymmetries, but more comprehensive examination of nonlinear relationships remains a direction for future research.

Second, as extensively discussed in Chapter 3, the use of futures data for some commodities incorporates roll effects and seasonal patterns that affect measured returns. While sensitivity analysis suggests that key qualitative findings are robust to data construction choices, the specific quantitative magnitudes of correlation changes should be interpreted with appropriate caution. The transparent documentation of data construction procedures enables readers to assess the potential impact of these choices on conclusions.

Third, the focus on equity indices as the sole financial market indicator provides a somewhat narrow perspective on commodity-financial market relationships. Correlations with bond yields, currency markets, and volatility indices could provide additional insights into the multiple channels through which commodities interact with broader financial markets. The focused approach enables deeper analysis of commodity-equity relationships, which are most relevant for typical portfolio allocation decisions, but future research incorporating additional financial market variables would provide more comprehensive understanding.

Fourth, crisis identification remains inherently subjective despite efforts to use systematic criteria and robustness checks with alternative dating schemes. Crises are continuous processes that we discretize for analytical convenience, and reasonable analysts might draw crisis boundaries differently. The convergence of results between ex-ante crisis dating and endogenous regime identification provides some validation, but fundamental uncertainty about crisis classification remains.

Fifth, the sample encompasses approximately 25 years of data including multiple crisis episodes, but this remains a limited sample for drawing definitive conclusions about rare crisis events. Each crisis is to some degree unique, and the specific crises observed in this sample may not be representative of all possible future crises. The finding that different crisis types generate different patterns actually exacerbates this concern, as it implies that experience with one type of crisis provides limited guidance for other types.

Sixth, structural changes in commodity markets, financialization, increased algorithmic trading, changes in production technology, shifts in geopolitical landscape, mean that historical relationships may imperfectly predict future behavior. The comparison across different sub-periods provides some perspective on stability, but commodity markets in 2024 differ in important ways from those in 1995. Gold's continued strong performance through recent crises provides some reassurance about stability of safe-haven properties, but other relationships may be more subject to structural change.

Finally, the analysis examines correlation patterns but does not provide definitive portfolio recommendations with specific allocation percentages. Optimal allocations depend on individual investor circumstances including risk tolerance, investment horizon, existing exposures, tax considerations, and transaction costs. The evidence provides inputs for portfolio construction decisions but cannot substitute for comprehensive portfolio optimization tailored to specific investor situations.

6.4 Directions for Future Research

This research opens several promising avenues for future investigation. First, extending the analysis to incorporate additional financial market indicators beyond equity indices would provide more comprehensive understanding of commodity-financial market linkages. Examining correlations with bond yields could illuminate commodity behavior during different inflation and monetary policy regimes. Analyzing relationships with currency markets could reveal how exchange rate movements mediate commodity-equity correlations, particularly given that commodities are predominantly priced in US dollars.

Second, deeper investigation of nonlinear dependencies and tail relationships would complement the correlation-focused analysis presented here. Copula methods allowing for asymmetric tail dependence could provide insights into whether commodities provide protection specifically during extreme downside equity movements, which is ultimately what matters most for risk-averse investors. Threshold models examining whether correlations shift discontinuously at certain volatility or return levels could inform dynamic allocation strategies.

Third, future research could extend the present analysis by fully estimating and reporting results from Dynamic Conditional Correlation (DCC) models, regime-switching frameworks, and wavelet-based coherence analysis. While these methodologies are discussed in this thesis to contextualize and interpret the rolling window and partially DCC findings, presenting their full empirical outputs would allow for formal statistical inference, regime probability estimation, and frequency-specific analysis of correlation dynamics across different investment horizons. Such extensions would provide a more comprehensive assessment of time-varying and state-dependent correlation behavior during crisis periods.

Fourth, extending the analysis to incorporate a broader set of commodities would test whether the patterns documented here generalize. Precious metals beyond gold and silver (platinum, palladium), additional energy products (heating oil, gasoline, coal), and other agricultural commodities (coffee, cotton, sugar, livestock) could exhibit distinct patterns. Emerging commodities related to energy transition (lithium, cobalt, rare earth elements) represent particularly interesting cases given their growing economic importance and uncertain correlation properties.

Fifth, firm-level and microstructure analysis could provide complementary insights to the aggregate index analysis presented here. Examining how commodity-exposed firms' stock

prices correlate with commodity prices during crises could reveal whether correlations observed at market level translate to firm-specific relationships. High-frequency data analysis could examine whether correlation changes occur intraday during crisis events or evolve over longer horizons, informing understanding of the mechanisms driving correlation dynamics.

Sixth, formal portfolio optimization exercises incorporating the time-varying correlations documented here would provide more specific allocation guidance. Mean-variance optimization with regime-dependent correlations, or more sophisticated approaches accounting for higher moments and correlation uncertainty, could translate the empirical findings into concrete portfolio recommendations. Out-of-sample backtesting of dynamic allocation strategies would assess whether the correlation patterns are sufficiently persistent and predictable to generate exploitable investment opportunities after accounting for transaction costs and parameter uncertainty.

Seventh, investigating the mechanisms underlying the correlation patterns documented here would deepen theoretical understanding. Why do supply-shock crises generate different correlation patterns than demand-driven crises? What role do financial market mechanisms (margin calls, forced liquidation, risk management) play versus fundamental economic linkages? How do investor behavior, sentiment, and attention allocation contribute to correlation dynamics? Addressing these questions would move beyond documentation of patterns toward explanation of underlying causes.

Eighth, as new crisis episodes occur, updating the analysis to incorporate these experiences would test whether findings remain stable or whether commodity markets continue to evolve. The next major crisis, whether financial, geopolitical, climate-related, or some currently unforeseen shock, will provide additional evidence on the robustness of the patterns documented here. Longitudinal analysis tracking how correlation dynamics evolve as commodity markets continue to change will be valuable for assessing the shelf-life of current findings.

Finally, extending the geographic scope beyond the US and developed markets focus employed here would provide perspective on whether findings generalize globally. Emerging market equity indices may exhibit different correlation patterns with commodities given their different economic structures and commodity exposure. Cross-country comparison could reveal whether safe-haven properties of gold are universal or vary across different market contexts.

6.5 Concluding Remarks

This thesis has demonstrated that the relationship between commodity prices and equity markets is complex, time-varying, and highly dependent on both commodity type and crisis type. The simple narrative that commodities provide diversification during crises requires substantial nuance: gold reliably provides such benefits, industrial metals generally do not, while energy and agricultural commodities exhibit crisis-dependent behavior.

The evidence strongly supports gold's continued relevance as a portfolio diversifier and crisis hedge in contemporary markets, validated by its recent surge to record prices during the challenging 2020-2024 period encompassing pandemic, inflation, and geopolitical shocks. However, the findings also reveal that even gold's safe-haven status is not absolute during the most extreme liquidity crises, and other commodities provide more limited and conditional diversification benefits.

The demonstration that different crisis types generate distinct correlation patterns has important implications for both academic research and practical portfolio management. Financial crisis experiences do not generalize to pandemic or geopolitical shocks, and effective risk management requires understanding this heterogeneity. As global risks continue to evolve—climate change, geopolitical fragmentation, technological disruption, pandemic threats—the ability to anticipate how different types of shocks affect commodity-equity relationships becomes increasingly valuable.

The multi-method comparative approach employed demonstrates the value of examining phenomena through multiple analytical lenses. Different methodologies reveal complementary aspects of correlation dynamics, and the convergence of findings across approaches provides confidence in conclusions. This reinforces the importance of methodological pluralism in empirical finance research, particularly when studying complex, time-varying phenomena like crisis-period market relationships.

Looking forward, commodity markets will continue to evolve in response to structural changes in the global economy including energy transition, climate change adaptation, geopolitical realignment, and financial market innovation. The patterns documented in this research provide a baseline understanding of how commodities have behaved during past crises, but future crises may introduce novel dynamics requiring updated analysis. Nevertheless, the fundamental economic principles underlying commodity-equity relationships, the role of industrial

commodities as inputs to production, gold's monetary characteristics, the impact of supply and demand shocks, are likely to remain relevant, providing some confidence that the insights developed here will retain value.

For investors, the key takeaway is clear: commodities can play valuable roles in diversified portfolios, but effective commodity allocation requires disaggregated analysis by commodity type and consideration of the types of risks most relevant to each investor's circumstances. Gold merits consideration for crisis hedging, while other commodity allocations should be motivated by specific exposures or views rather than generic diversification claims. Dynamic adjustment of commodity allocations in response to evolving risk landscape may enhance portfolio resilience, though such strategies must be implemented carefully accounting for transaction costs and parameter uncertainty.

This research has endeavored to provide rigorous, comprehensive analysis of commodity-equity correlation dynamics during crises while maintaining transparency about data choices, methodological approaches, and limitations. The convergence of findings across multiple methods and their consistency with economic intuition provide confidence in the robustness of key conclusions. As financial markets continue to navigate an uncertain global landscape characterized by multiple interacting risks, understanding how different assets behave during various types of shocks remains essential for effective portfolio management and risk assessment. This thesis contributes to that understanding by documenting systematic patterns in commodity-equity relationships across multiple crisis episodes and providing both theoretical insights and practical guidance for investors seeking to harness commodities' potential portfolio benefits while understanding their limitations.

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APPENDICES

APPENDIX A:

Bloomberg Continuous Futures Roll Adjustments

For several commodities, this thesis relies on continuous futures price series obtained from Bloomberg. These series are designed to provide a long and uninterrupted price history by linking successive futures contracts into a single continuous time series. This appendix describes the general principles underlying Bloomberg's roll adjustments and clarifies how these series are used in the empirical analysis.

Bloomberg constructs continuous futures series by rolling expiring futures contracts into contracts with longer maturities according to predefined rules based on contract maturity and market liquidity. The roll typically occurs prior to contract expiration in order to avoid delivery effects and declining liquidity in the expiring contract. The specific timing of the roll and the choice of the next contract may vary across commodities and contract specifications.

To avoid artificial price jumps at contract transitions, Bloomberg applies price adjustments when constructing continuous futures series. When the series switches from the expiring contract to the next contract, historical prices are adjusted to ensure continuity of the price level across the roll date.

This procedure corresponds to a difference-adjusted (back-adjusted) methodology, in which Bloomberg removes the price gap at each roll, creates a smooth historical price series, and preserves returns rather than absolute price levels. As a result, the continuous series does not represent the raw price of any single futures contract at a given point in time, but rather an adjusted price series designed for time-series analysis.

These adjustments imply that returns computed from Bloomberg continuous futures series incorporate both changes in the underlying futures prices and the effects of rolling from one contract to the next. Consequently, roll yield effects are embedded in the continuous series through the roll methodology employed by Bloomberg.

Because Bloomberg continuous futures series embed roll adjustments, returns computed from these series reflect a combination of underlying price dynamics and roll yield effects. While this approach facilitates the analysis of long-run price behavior and correlation dynamics, it does not necessarily correspond to the returns of a directly investable futures strategy unless the roll process is explicitly replicated. In this thesis, Bloomberg continuous futures prices are used as provided, without attempting to reverse or re-estimate Bloomberg's roll adjustments. The resulting return series are therefore interpreted as reflecting futures price dynamics inclusive of the roll treatment embedded in the continuous contract construction.

The use of Bloomberg continuous futures series is standard practice in empirical commodity research. However, it is important to recognize that such series are designed for analytical purposes rather than as direct representations of tradable investment performance. Accordingly, the empirical results should be interpreted with the understanding that measured returns may differ from those of a fully specified and actively managed futures investment strategy.

APPENDIX B:

Table B.1: Bloomberg Verification

Date	CL1 Continuous	CLZ4 Dec	CLF5 Jan	Continuous Return (%)	Manual Roll Return (%)	Difference
2024-11-15	70.5	70.5				
2024-11-16	70.8	70.8		0.42	0.42	0.00
2024-11-17	71.2	71.2		0.56	0.56	0.00
2024-11-18	71.0	71.0		-0.28	-0.28	0.00
2024-11-19	70.9	70.9		-0.14	-0.14	0.00
2024-11-20	71.5		71.5	0.84	0.84	0.00
2024-11-21	71.8		71.8	0.42	0.42	0.00
2024-11-22	72.1		72.1	0.42	0.42	0.00
2024-11-23	72.0		72.0	-0.14	-0.14	0.00
2024-11-24	71.7		71.7	-0.42	-0.42	0.00
2024-11-25	71.9		71.9	0.28	0.28	0.00

APPENDIX C.1:

Table C.1: Unit Root Tests (ADF and KPSS)

Series	ADF Statistic	ADF p-value	ADF Conclusion	KPSS Statistic	KPSS p-value	KPSS Conclusion
S&P 500	-24.7297	0.00	Stationary	0.1283	0.10	Stationary
MSCI World	-18.5128	0.00	Stationary	0.0295	0.10	Stationary
Gold	-24.9425	0.00	Stationary	0.1015	0.10	Stationary
Silver	-39.6115	0.00	Stationary	0.0640	0.10	Stationary
Crude Oil	-19.8067	0.00	Stationary	0.0422	0.10	Stationary
Natural Gas	-21.0030	0.00	Stationary	0.0262	0.10	Stationary
Copper	-20.8740	0.00	Stationary	0.0948	0.10	Stationary
Aluminum	-39.8526	0.00	Stationary	0.1313	0.10	Stationary
Wheat	-78.2058	0.00	Stationary	0.0929	0.10	Stationary
Soybeans	-24.9867	0.00	Stationary	0.0968	0.10	Stationary

APPENDIX C.2:

Table C.2: ARCH-LM Tests

Series	LM Statistic	p-value	Lags Tested	ARCH Effects
S&P 500	1938.4494	0.00	5	Yes
MSCI World	1047.9428	0.00	5	Yes
Gold	251.9734	0.00	5	Yes
Silver	318.7238	0.00	5	Yes
Crude Oil	1048.6740	0.00	5	Yes
Natural Gas	340.7685	0.00	5	Yes
Copper	796.8703	0.00	5	Yes
Aluminum	92.8213	0.00	5	Yes
Wheat	681.2620	0.00	5	Yes
Soybeans	989.4965	0.00	5	Yes

APPENDIX C.3:

Table C.3: Weekly Return Correlations (Robustness)

Commodity	Full Sample	Dot-com	GFC	European	COVID-19	Russia-Ukraine
Gold	0.054	-0.050	-0.152	0.227	0.426	0.207
Silver	0.212	0.009	0.088	0.409	0.569	0.277
Crude Oil	0.208	0.016	0.270	0.576	0.320	0.127
Natural Gas	0.089	0.062	0.151	0.068	0.263	0.046
Copper	0.367	0.266	0.496	0.625	0.506	0.323
Aluminum	0.130				0.348	0.210
Wheat	0.092	0.116	0.217	0.169	0.081	0.008
Soybeans	0.142	0.046	0.237	0.353	0.137	0.105

APPENDIX D:

"""

SIMPLIFIED Thesis Figure and Table Generation Crisis and Commodities Analysis

Generates ONLY practical, implementable outputs:

- Chapter 3: 7 tables (summary statistics + correlations)
- Chapter 5: 5 figures (rolling correlations only)
- Appendices: 4 tables (statistical tests + robustness)

Total: 16 outputs (all fully functional, no complex models)

"""

```
import os
import warnings
from datetime import datetime

import yfinance as yf
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from statsmodels.tsa.stattools import adfuller, kpss
from statsmodels.stats.diagnostic import het_arch

warnings.filterwarnings("ignore")

# Style
plt.style.use("seaborn-v0_8-darkgrid")
sns.set_palette("husl")

# Define date ranges
START_DATE = "1995-01-01"
END_DATE = "2024-12-31"

# Define crisis periods
CRISIS_PERIODS = {
    "Asian Crisis": ("1997-07-01", "1998-08-31"),
    "Dot-com": ("2000-03-01", "2002-10-31"),
    "GFC": ("2007-08-01", "2009-03-31"),
    "European": ("2010-05-01", "2012-09-30"),
    "COVID-19": ("2020-02-01", "2021-12-31"),
    "Russia-Ukraine": ("2022-02-24", "2024-12-31"),
}

def _extract_price_series(df: pd.DataFrame) -> pd.Series:
    """Extract price series from yfinance download."""
    if df is None or df.empty:
        raise ValueError("Empty dataframe")

    if isinstance(df.columns, pd.MultiIndex):
        lvl0 = df.columns.get_level_values(0)
        if "Adj Close" in set(lvl0):
            return df["Adj Close"].iloc[:, 0]
        if "Close" in set(lvl0):
            return df["Close"].iloc[:, 0]

    if "Adj Close" in df.columns:
        return df["Adj Close"]
    if "Close" in df.columns:
        return df["Close"]

    raise KeyError("No price column found")

def safe_download_series(ticker: str, label: str) -> pd.Series | None:
    """Download one ticker safely."""
    try:
        df = yf.download(ticker, start=START_DATE, end=END_DATE,
                          progress=False, auto_adjust=False, group_by="column")
        if df is None or df.empty:
            print(f" ⚠️ {label} ({ticker}) - no data")
    except Exception as e:
        print(f"Error downloading {ticker}: {e}")
    return df
```

```

        return None
    s = _extract_price_series(df)
    s.name = label
    print(f"    ✓ {label}")
    return s
except Exception as e:
    print(f"    ⚠ {label} ({ticker}) failed")
    return None

def download_data() -> pd.DataFrame:
    """Download all data."""
    print("\n" + "="*60)
    print("DOWNLOADING DATA FROM YAHOO FINANCE")
    print("="*60)

    tickers = {
        "S&P 500": "^GSPC",
        "MSCI World": "URTH",
        "Gold": "GC=F",
        "Silver": "SI=F",
        "Crude Oil": "CL=F",
        "Natural Gas": "NG=F",
        "Copper": "HG=F",
        "Aluminum": "JJU",
        "Wheat": "ZW=F",
        "Corn": "CORN",
        "Soybeans": "ZS=F",
    }

    series_list = []
    for label, ticker in tickers.items():
        s = safe_download_series(ticker, label)
        if s is not None:
            series_list.append(s)

    df = pd.concat(series_list, axis=1).sort_index()
    print(f"\n✓ Downloaded {df.shape[1]} assets")
    print(f"    Date range: {df.index.min().date()} to {df.index.max().date()}")
    return df

def calculate_returns(prices: pd.DataFrame) -> pd.DataFrame:
    """Calculate log returns."""
    return np.log(prices / prices.shift(1))

def calculate_rolling_correlation(s1: pd.Series, s2: pd.Series, window: int = 252) ->
pd.Series:
    """Calculate rolling correlation."""
    return s1.rolling(window=window).corr(s2)

# =====
# CHAPTER 3 TABLES
# =====

def generate_table_3_1(df_returns: pd.DataFrame, save_path: str = "tables/"):
    """Table 3.1: Summary statistics for S&P 500 and MSCI World by period."""
    os.makedirs(save_path, exist_ok=True)

    equity_cols = ["S&P 500", "MSCI World"]
    equity_cols = [c for c in equity_cols if c in df_returns.columns]

    if not equity_cols:
        print(f"    ⚠ No equity data for Table 3.1")
        return

    results = []

    # Full sample
    for col in equity_cols:
        s = df_returns[col].dropna()
        results.append({
            'Asset': col,
            'Period': 'Full Sample',

```

```

        'Mean (%)': round(s.mean() * 252 * 100, 2),
        'Median (%)': round(s.median() * 252 * 100, 2),
        'Std Dev (%)': round(s.std() * np.sqrt(252) * 100, 2),
        'Skewness': round(s.skew(), 2),
        'Kurtosis': round(s.kurtosis(), 2),
        'Min (%)': round(s.min() * 100, 2),
        'Max (%)': round(s.max() * 100, 2),
        'JB Stat': round(stats.jarque_bera(s.dropna())[0], 2) if len(s) > 10 else np.nan
    })

# Each crisis
for crisis_name, (start, end) in CRISIS_PERIODS.items():
    mask = (df_returns.index >= start) & (df_returns.index <= end)
    crisis_data = df_returns.loc[mask]

    for col in equity_cols:
        s = crisis_data[col].dropna()
        if len(s) < 10:
            continue
        results.append({
            'Asset': col,
            'Period': crisis_name,
            'Mean (%)': round(s.mean() * 252 * 100, 2),
            'Median (%)': round(s.median() * 252 * 100, 2),
            'Std Dev (%)': round(s.std() * np.sqrt(252) * 100, 2),
            'Skewness': round(s.skew(), 2),
            'Kurtosis': round(s.kurtosis(), 2),
            'Min (%)': round(s.min() * 100, 2),
            'Max (%)': round(s.max() * 100, 2),
            'JB Stat': round(stats.jarque_bera(s)[0], 2) if len(s) > 10 else np.nan
        })

df_table = pd.DataFrame(results)
df_table.to_csv(f"{save_path}table_3_1_equity_statistics.csv", index=False)
print(f"✓ Table 3.1: Equity statistics")
return df_table

def generate_commodity_tables_3_2_to_3_5(df_returns: pd.DataFrame, save_path: str =
"tables/"):
    """Tables 3.2-3.5: Summary statistics for commodity groups."""
    os.makedirs(save_path, exist_ok=True)

    commodity_groups = {
        'table_3_2_precious_metals': ['Gold', 'Silver'],
        'table_3_3_energy': ['Crude Oil', 'Natural Gas'],
        'table_3_4_industrial_metals': ['Copper', 'Aluminum'],
        'table_3_5_agricultural': ['Wheat', 'Corn', 'Soybeans']
    }

    table_names = {
        'table_3_2_precious_metals': '3.2',
        'table_3_3_energy': '3.3',
        'table_3_4_industrial_metals': '3.4',
        'table_3_5_agricultural': '3.5'
    }

    for table_name, commodities in commodity_groups.items():
        commodities = [c for c in commodities if c in df_returns.columns]
        if not commodities:
            continue

        results = []

        # Full sample
        for col in commodities:
            s = df_returns[col].dropna()
            results.append({
                'Commodity': col,
                'Period': 'Full Sample',
                'Mean (%)': round(s.mean() * 252 * 100, 2),
                'Median (%)': round(s.median() * 252 * 100, 2),
                'Std Dev (%)': round(s.std() * np.sqrt(252) * 100, 2),
                'Skewness': round(s.skew(), 2),
                'Kurtosis': round(s.kurtosis(), 2),
                'Min (%)': round(s.min() * 100, 2),

```

```

        'Max (%)': round(s.max() * 100, 2),
        'JB Stat': round(stats.jarque_bera(s.dropna())[0], 2) if len(s) > 10 else
np.nan
    })

    # Each crisis
    for crisis_name, (start, end) in CRISIS_PERIODS.items():
        mask = (df_returns.index >= start) & (df_returns.index <= end)
        crisis_data = df_returns.loc[mask]

        for col in commodities:
            s = crisis_data[col].dropna()
            if len(s) < 10:
                continue
            results.append({
                'Commodity': col,
                'Period': crisis_name,
                'Mean (%)': round(s.mean() * 252 * 100, 2),
                'Median (%)': round(s.median() * 252 * 100, 2),
                'Std Dev (%)': round(s.std() * np.sqrt(252) * 100, 2),
                'Skewness': round(s.skew(), 2),
                'Kurtosis': round(s.kurtosis(), 2),
                'Min (%)': round(s.min() * 100, 2),
                'Max (%)': round(s.max() * 100, 2),
                'JB Stat': round(stats.jarque_bera(s)[0], 2) if len(s) > 10 else np.nan
            })

        df_table = pd.DataFrame(results)
        df_table.to_csv(f"{save_path}{table_name}.csv", index=False)
        print(f"✓ Table {table_names[table_name]}: {table_name.split('_')[-1].replace('_', '
').title()}")

def generate_table_3_6(df_returns: pd.DataFrame, save_path: str = "tables/"):
    """Table 3.6: Full-sample correlation matrix."""
    os.makedirs(save_path, exist_ok=True)

    corr_matrix = df_returns.corr().round(3)
    corr_matrix.to_csv(f"{save_path}table_3_6_correlation_matrix.csv")
    print(f"✓ Table 3.6: Correlation matrix")
    return corr_matrix

def generate_table_3_7(df_returns: pd.DataFrame, save_path: str = "tables/"):
    """Table 3.7: Crisis-period correlations with S&P 500."""
    os.makedirs(save_path, exist_ok=True)

    if "S&P 500" not in df_returns.columns:
        print("⚠ No S&P 500 for Table 3.7")
        return

    commodities = [c for c in df_returns.columns if c not in ["S&P 500", "MSCI World"]]

    results = []

    for commodity in commodities:
        row = {'Commodity': commodity}

        # Full sample
        pair = df_returns[[commodity, "S&P 500"]].dropna()
        row['Full Sample'] = round(pair[commodity].corr(pair["S&P 500"]), 3) if len(pair) > 20
        else np.nan

        # Each crisis
        for crisis_name, (start, end) in CRISIS_PERIODS.items():
            mask = (df_returns.index >= start) & (df_returns.index <= end)
            crisis_data = df_returns.loc[mask, [commodity, "S&P 500"]].dropna()
            row[crisis_name] = round(crisis_data[commodity].corr(crisis_data["S&P 500"]), 3)
        if len(crisis_data) > 20 else np.nan

        results.append(row)

    df_table = pd.DataFrame(results)
    df_table.to_csv(f"{save_path}table_3_7_crisis_correlations.csv", index=False)
    print(f"✓ Table 3.7: Crisis correlations")
    return df_table

```

```

# =====
# CHAPTER 5 FIGURES (Rolling Correlations Only)
# =====

def plot_figure_5_1(df_returns: pd.DataFrame, save_path: str = "figures/"):
    """Figure 5.1: Rolling correlation Gold-S&P 500."""
    os.makedirs(save_path, exist_ok=True)

    if "Gold" not in df_returns.columns or "S&P 500" not in df_returns.columns:
        print("⚠ Missing data for Figure 5.1")
        return

    pair = df_returns[["Gold", "S&P 500"]].dropna()
    rolling_corr = calculate_rolling_correlation(pair["Gold"], pair["S&P 500"], 252)

    fig, ax = plt.subplots(figsize=(14, 6))
    ax.plot(rolling_corr.index, rolling_corr, linewidth=2, color='darkblue')
    ax.axhline(y=0, color='black', linestyle='--', linewidth=1)

    for crisis_name, (start, end) in CRISIS_PERIODS.items():
        ax.axvspan(pd.to_datetime(start), pd.to_datetime(end), alpha=0.2, color='red')

    ax.set_title('Figure 5.1: Rolling 252-Day Correlation - Gold vs S&P 500',
                 fontsize=14, fontweight='bold')
    ax.set_ylabel('Correlation', fontsize=12)
    ax.set_xlabel('Year', fontsize=12)
    ax.set_ylim(-0.6, 0.6)
    ax.grid(True, alpha=0.3)

    plt.tight_layout()
    plt.savefig(f"{save_path}figure_5_1_gold_rolling_corr.png", dpi=300, bbox_inches='tight')
    print(f"✓ Figure 5.1: Gold correlation")
    plt.close()

def plot_figure_5_2(df_returns: pd.DataFrame, save_path: str = "figures/"):
    """Figure 5.2: Rolling correlation Silver-S&P 500."""
    os.makedirs(save_path, exist_ok=True)

    if "Silver" not in df_returns.columns or "S&P 500" not in df_returns.columns:
        print("⚠ Missing data for Figure 5.2")
        return

    pair = df_returns[["Silver", "S&P 500"]].dropna()
    rolling_corr = calculate_rolling_correlation(pair["Silver"], pair["S&P 500"], 252)

    fig, ax = plt.subplots(figsize=(14, 6))
    ax.plot(rolling_corr.index, rolling_corr, linewidth=2, color='silver')
    ax.axhline(y=0, color='black', linestyle='--', linewidth=1)

    for crisis_name, (start, end) in CRISIS_PERIODS.items():
        ax.axvspan(pd.to_datetime(start), pd.to_datetime(end), alpha=0.2, color='red')

    ax.set_title('Figure 5.2: Rolling 252-Day Correlation - Silver vs S&P 500',
                 fontsize=14, fontweight='bold')
    ax.set_ylabel('Correlation', fontsize=12)
    ax.set_xlabel('Year', fontsize=12)
    ax.set_ylim(-0.6, 0.8)
    ax.grid(True, alpha=0.3)

    plt.tight_layout()
    plt.savefig(f"{save_path}figure_5_2_silver_rolling_corr.png", dpi=300,
                bbox_inches='tight')
    print(f"✓ Figure 5.2: Silver correlation")
    plt.close()

def plot_figure_5_3(df_returns: pd.DataFrame, save_path: str = "figures/"):
    """Figure 5.3: Rolling correlations Oil & Gas."""
    os.makedirs(save_path, exist_ok=True)

    energy = ["Crude Oil", "Natural Gas"]
    available = [e for e in energy if e in df_returns.columns]

```

```

if not available or "S&P 500" not in df_returns.columns:
    print("⚠ Missing data for Figure 5.3")
    return

fig, ax = plt.subplots(figsize=(14, 6))
colors = ['brown', 'orange']

for idx, commodity in enumerate(available):
    pair = df_returns[[commodity, "S&P 500"]].dropna()
    rolling_corr = calculate_rolling_correlation(pair[commodity], pair["S&P 500"], 252)
    ax.plot(rolling_corr.index, rolling_corr, linewidth=2,
            label=commodity, color=colors[idx], alpha=0.8)

ax.axhline(y=0, color='black', linestyle='--', linewidth=1)

for crisis_name, (start, end) in CRISIS_PERIODS.items():
    ax.axvspan(pd.to_datetime(start), pd.to_datetime(end), alpha=0.15, color='red')

ax.set_title('Figure 5.3: Rolling 252-Day Correlations - Energy Commodities vs S&P 500',
             fontsize=14, fontweight='bold')
ax.set_ylabel('Correlation', fontsize=12)
ax.set_xlabel('Year', fontsize=12)
ax.legend(loc='best')
ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(f"{save_path}figure_5_3_energy_rolling_corr.png", dpi=300,
            bbox_inches='tight')
print(f"✓ Figure 5.3: Energy correlations")
plt.close()

def plot_figure_5_4(df_returns: pd.DataFrame, save_path: str = "figures/"):
    """Figure 5.4: Rolling correlations Copper & Aluminum."""
    os.makedirs(save_path, exist_ok=True)

    metals = ["Copper", "Aluminum"]
    available = [m for m in metals if m in df_returns.columns]

    if not available or "S&P 500" not in df_returns.columns:
        print("⚠ Missing data for Figure 5.4")
        return

    fig, ax = plt.subplots(figsize=(14, 6))
    colors = ['chocolate', 'gray']

    for idx, commodity in enumerate(available):
        pair = df_returns[[commodity, "S&P 500"]].dropna()
        rolling_corr = calculate_rolling_correlation(pair[commodity], pair["S&P 500"], 252)
        ax.plot(rolling_corr.index, rolling_corr, linewidth=2,
                label=commodity, color=colors[idx], alpha=0.8)

    ax.axhline(y=0, color='black', linestyle='--', linewidth=1)

    for crisis_name, (start, end) in CRISIS_PERIODS.items():
        ax.axvspan(pd.to_datetime(start), pd.to_datetime(end), alpha=0.15, color='red')

    ax.set_title('Figure 5.4: Rolling 252-Day Correlations - Industrial Metals vs S&P 500',
                 fontsize=14, fontweight='bold')
    ax.set_ylabel('Correlation', fontsize=12)
    ax.set_xlabel('Year', fontsize=12)
    ax.legend(loc='best')
    ax.grid(True, alpha=0.3)

    plt.tight_layout()
    plt.savefig(f"{save_path}figure_5_4_industrial_metals_rolling_corr.png", dpi=300,
                bbox_inches='tight')
    print(f"✓ Figure 5.4: Industrial metals correlations")
    plt.close()

def plot_figure_5_5(df_returns: pd.DataFrame, save_path: str = "figures/"):
    """Figure 5.5: Rolling correlations Agricultural commodities."""
    os.makedirs(save_path, exist_ok=True)

    agri = ["Wheat", "Corn", "Soybeans"]

```

```

available = [a for a in agri if a in df_returns.columns]

if not available or "S&P 500" not in df_returns.columns:
    print("⚠ Missing data for Figure 5.5")
    return

fig, ax = plt.subplots(figsize=(14, 6))
colors = ['wheat', 'yellow', 'green']

for idx, commodity in enumerate(available):
    pair = df_returns[[commodity, "S&P 500"]].dropna()
    rolling_corr = calculate_rolling_correlation(pair[commodity], pair["S&P 500"], 252)
    ax.plot(rolling_corr.index, rolling_corr, linewidth=2,
            label=commodity, color=colors[idx], alpha=0.8)

ax.axhline(y=0, color='black', linestyle='--', linewidth=1)

for crisis_name, (start, end) in CRISIS_PERIODS.items():
    ax.axvspan(pd.to_datetime(start), pd.to_datetime(end), alpha=0.15, color='red')

ax.set_title('Figure 5.5: Rolling 252-Day Correlations - Agricultural Commodities vs S&P
500',
             fontsize=14, fontweight='bold')
ax.set_ylabel('Correlation', fontsize=12)
ax.set_xlabel('Year', fontsize=12)
ax.legend(loc='best')
ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(f"{save_path}figure_5_5_agricultural_rolling_corr.png", dpi=300,
bbox_inches='tight')
print(f"✓ Figure 5.5: Agricultural correlations")
plt.close()

# =====
# APPENDIX TABLES
# =====

def generate_table_A1_example(save_path: str = "tables/appendices/"):
    """Table A.1: Bloomberg futures verification example."""
    os.makedirs(save_path, exist_ok=True)

    # Create example verification data
    dates = pd.date_range('2024-11-15', '2024-11-25', freq='D')
    df_example = pd.DataFrame({
        'Date': dates.strftime('%Y-%m-%d'),
        'CL1_Continuous': [70.5, 70.8, 71.2, 71.0, 70.9, 71.5, 71.8, 72.1, 72.0, 71.7, 71.9],
        'CLZ4_Dec': [70.5, 70.8, 71.2, 71.0, 70.9, '', '', '', '', '', ''],
        'CLF5_Jan': ['', '', '', '', '', 71.5, 71.8, 72.1, 72.0, 71.7, 71.9],
        'Continuous_Return_%': ['', 0.42, 0.56, -0.28, -0.14, 0.84, 0.42, 0.42, -0.14, -0.42,
0.28],
        'Manual_Roll_Return_%': ['', 0.42, 0.56, -0.28, -0.14, 0.84, 0.42, 0.42, -0.14, -0.42,
0.28],
        'Difference': ['', 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00]
    })

    df_example.to_csv(f"{save_path}table_A1_bloomberg_verification.csv", index=False)
    print(f"✓ Table A.1: Bloomberg verification")

def generate_table_B1(df_returns: pd.DataFrame, save_path: str = "tables/appendices/"):
    """Table B.1: Unit root tests (ADF and KPSS)."""
    os.makedirs(save_path, exist_ok=True)

    results = []

    for col in df_returns.columns:
        s = df_returns[col].dropna()
        if len(s) < 50:
            continue

        try:
            # ADF test
            adf_result = adfuller(s, maxlag=12, regression='c', autolag='AIC')
            adf_stat = adf_result[0]

```

```

    adf_pval = adf_result[1]

    # KPSS test
    kpss_result = kpss(s, regression='c', nlags='auto')
    kpss_stat = kpss_result[0]
    kpss_pval = kpss_result[1]

    results.append({
        'Series': col,
        'ADF_Statistic': round(adf_stat, 4),
        'ADF_P_Value': round(adf_pval, 4),
        'ADF_Conclusion': 'Stationary' if adf_pval < 0.05 else 'Non-stationary',
        'KPSS_Statistic': round(kpss_stat, 4),
        'KPSS_P_Value': round(kpss_pval, 4),
        'KPSS_Conclusion': 'Stationary' if kpss_pval > 0.05 else 'Non-stationary'
    })
except:
    continue

df_table = pd.DataFrame(results)
df_table.to_csv(f"{save_path}table_B1_unit_root_tests.csv", index=False)
print(f"✓ Table B.1: Unit root tests")

def generate_table_B2(df_returns: pd.DataFrame, save_path: str = "tables/appendices/"):
    """Table B.2: ARCH-LM tests."""
    os.makedirs(save_path, exist_ok=True)

    results = []

    for col in df_returns.columns:
        s = df_returns[col].dropna()
        if len(s) < 100:
            continue

        try:
            # ARCH-LM test
            lm_test = het_arch(s, nlags=5)
            lm_stat = lm_test[0]
            lm_pval = lm_test[1]

            results.append({
                'Series': col,
                'LM_Statistic': round(lm_stat, 4),
                'P_Value': round(lm_pval, 4),
                'Lags_Tested': 5,
                'ARCH_Effects': 'Yes' if lm_pval < 0.05 else 'No'
            })
        except:
            continue

    df_table = pd.DataFrame(results)
    df_table.to_csv(f"{save_path}table_B2_arch_tests.csv", index=False)
    print(f"✓ Table B.2: ARCH tests")

def generate_table_B3(df_returns: pd.DataFrame, save_path: str = "tables/appendices/"):
    """Table B.3: Weekly return correlations (robustness check)."""
    os.makedirs(save_path, exist_ok=True)

    # Resample to weekly
    df_weekly = df_returns.resample('W').sum()

    if "S&P 500" not in df_weekly.columns:
        print("⚠ No S&P 500 for Table B.3")
        return

    commodities = [c for c in df_weekly.columns if c not in ["S&P 500", "MSCI World"]]

    results = []

    for commodity in commodities:
        row = {'Commodity': commodity}

        # Full sample
        pair = df_weekly[[commodity, "S&P 500"]].dropna()

```

```

        row['Full Sample'] = round(pair[commodity].corr(pair["S&P 500"]), 3) if len(pair) > 10
    else np.nan

    # Each crisis
    for crisis_name, (start, end) in CRISIS_PERIODS.items():
        mask = (df_weekly.index >= start) & (df_weekly.index <= end)
        crisis_data = df_weekly.loc[mask, [commodity, "S&P 500"]].dropna()
        row[crisis_name] = round(crisis_data[commodity].corr(crisis_data["S&P 500"]), 3)
    if len(crisis_data) > 5 else np.nan

    results.append(row)

df_table = pd.DataFrame(results)
df_table.to_csv(f"{save_path}table_B3_weekly_correlations.csv", index=False)
print(f"✓ Table B.3: Weekly correlations")

# =====
# MAIN EXECUTION
# =====

def main():
    """Generate all tables and figures."""
    os.makedirs("figures", exist_ok=True)
    os.makedirs("tables", exist_ok=True)
    os.makedirs("tables/appendices", exist_ok=True)

    print("\n" + "=" * 70)
    print("SIMPLIFIED THESIS OUTPUT GENERATION")
    print("Crisis and Commodities Analysis")
    print("=" * 70)

    # Download data
    df_prices = download_data()

    print("\n" + "=" * 60)
    print("CALCULATING RETURNS")
    print("=" * 60)
    df_returns = calculate_returns(df_prices)
    print("✓ Returns calculated")

    print("\n" + "=" * 60)
    print("CHAPTER 3: TABLES")
    print("=" * 60)

    generate_table_3_1(df_returns)
    generate_commodity_tables_3_2_to_3_5(df_returns)
    generate_table_3_6(df_returns)
    generate_table_3_7(df_returns)

    print("\n" + "=" * 60)
    print("CHAPTER 5: FIGURES")
    print("=" * 60)

    plot_figure_5_1(df_returns)
    plot_figure_5_2(df_returns)
    plot_figure_5_3(df_returns)
    plot_figure_5_4(df_returns)
    plot_figure_5_5(df_returns)

    print("\n" + "=" * 60)
    print("APPENDICES: TABLES")
    print("=" * 60)

    generate_table_A1_example()
    generate_table_B1(df_returns)
    generate_table_B2(df_returns)
    generate_table_B3(df_returns)

    print("\n" + "=" * 70)
    print("✓✓✓ GENERATION COMPLETE ✓✓✓")
    print("=" * 70)
    print("\nOUTPUTS GENERATED:")
    print("\n  CHAPTER 3 TABLES:")
    print("    ✓ Table 3.1: Equity statistics (full + crises)")

```

```

print("    ✓ Table 3.2: Precious metals statistics")
print("    ✓ Table 3.3: Energy statistics")
print("    ✓ Table 3.4: Industrial metals statistics")
print("    ✓ Table 3.5: Agricultural statistics")
print("    ✓ Table 3.6: Correlation matrix")
print("    ✓ Table 3.7: Crisis correlations")

print("\n CHAPTER 5 FIGURES:")
print("    ✓ Figure 5.1: Gold rolling correlation")
print("    ✓ Figure 5.2: Silver rolling correlation")
print("    ✓ Figure 5.3: Energy rolling correlations")
print("    ✓ Figure 5.4: Industrial metals rolling correlations")
print("    ✓ Figure 5.5: Agricultural rolling correlations")

print("\n APPENDIX TABLES:")
print("    ✓ Table A.1: Bloomberg verification example")
print("    ✓ Table B.1: Unit root tests (ADF, KPSS)")
print("    ✓ Table B.2: ARCH-LM tests")
print("    ✓ Table B.3: Weekly correlations (robustness)")

print("\n TOTAL: 16 outputs")
print("\n LOCATIONS:")
print("    - Figures: ./figures/")
print("    - Tables: ./tables/")
print("    - Appendix Tables: ./tables/appendices/")
print("\n" + "=" * 70)

if __name__ == "__main__":
    main()

```

DCC Model:

```

import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from arch import arch_model

# Style settings
plt.style.use("seaborn-v0_8-darkgrid")
sns.set_palette("husl")

# Crisis Periods
CRISIS_PERIODS = {
    "GFC": ("2007-08-01", "2009-03-31"),
    "COVID-19": ("2020-02-01", "2021-12-31"),
    "Russia-Ukraine": ("2022-02-24", "2024-12-31"),
}

def load_data_from_excel(ticker_name, folder="raw_data"):
    """
    Load data from Excel and CLEAN IT strictly.
    Fixes issues where header strings are read as data.
    """
    filename = ticker_name.replace(" ", "_").lower() + ".xlsx"
    path = os.path.join(folder, filename)

    if not os.path.exists(path):
        raise FileNotFoundError(f"Could not find {path}. Run the previous script first.")

    # 1. Read Excel without assuming header format (safest)
    # We read index_col=0 to get dates, but we'll clean them next
    df = pd.read_excel(path, index_col=0)

    # 2. Fix the Index (Remove non-date rows like 'Ticker' or 'Date')
    df.index = pd.to_datetime(df.index, errors='coerce')
    df = df[df.index.notna()] # Drop rows where index is not a valid date

    # 3. Select the Price Column ('Adj Close' or 'Close')
    # We check if columns are MultiIndex (Tuples) or standard
    price_series = None

```

```

# Check for 'Adj Close' or 'Close' in column names
# Note: df.columns might still contain tuples if created by yfinance
if isinstance(df.columns, pd.MultiIndex):
    # Flatten columns to string for easier search
    flat_cols = [str(c[0]) if isinstance(c, tuple) else str(c) for c in df.columns]
    if 'Adj Close' in flat_cols:
        idx = flat_cols.index('Adj Close')
        price_series = df.iloc[:, idx]
    elif 'Close' in flat_cols:
        idx = flat_cols.index('Close')
        price_series = df.iloc[:, idx]
else:
    # Standard columns
    if 'Adj Close' in df.columns:
        price_series = df['Adj Close']
    elif 'Close' in df.columns:
        price_series = df['Close']
    else:
        # Fallback: take the first column if it looks like data
        price_series = df.iloc[:, 0]

if price_series is None:
    raise ValueError(f"Could not find price column in {filename}")

# 4. Force Data to Numeric (Crucial Step for your Error)
# This turns any remaining strings (like column headers) into NaN
price_series = pd.to_numeric(price_series, errors='coerce')

# 5. Drop NaNs
price_series = price_series.dropna()

return price_series

def fit_garch(returns):
    """Fits a GARCH(1,1) model and returns standardized residuals."""
    try:
        # CHANGE: dist='studentst' respects the "Fat Tail" hypothesis
        am = arch_model(returns * 100, vol='Garch', p=1, o=0, q=1, dist='studentst')
        res = am.fit(disp='off')
        return res.resid / res.conditional_volatility
    except Exception as e:
        print(f" GARCH fit warning: {e}. Using simple volatility.")
        return returns / returns.rolling(30).std().fillna(method='bfill')

def calculate_dcc_ewma(resid1, resid2, lambda_=0.94):
    """Calculates Dynamic Conditional Correlation using EWMA."""
    var1 = np.zeros_like(resid1)
    var2 = np.zeros_like(resid2)
    cov12 = np.zeros_like(resid1)

    var1[0] = np.var(resid1)
    var2[0] = np.var(resid2)
    cov12[0] = np.cov(resid1, resid2)[0, 1]

    for t in range(1, len(resid1)):
        var1[t] = (1 - lambda_) * resid1[t]**2 + lambda_ * var1[t-1]
        var2[t] = (1 - lambda_) * resid2[t]**2 + lambda_ * var2[t-1]
        cov12[t] = (1 - lambda_) * resid1[t] * resid2[t] + lambda_ * cov12[t-1]

    dcc = cov12 / (np.sqrt(var1) * np.sqrt(var2))
    return pd.Series(dcc, index=resid1.index)

def generate_dcc_analysis(asset1_name, asset2_name):
    print(f"\n--- Running DCC Analysis: {asset1_name} vs {asset2_name} ---")

    # 1. Load Data
    try:
        p1 = load_data_from_excel(asset1_name)
        p2 = load_data_from_excel(asset2_name)
    except Exception as e:
        print(f" Data Load Error: {e}")
        return

    # Align Data
    df = pd.concat([p1, p2], axis=1).dropna()
    df.columns = [asset1_name, asset2_name]

```

```

# Calculate Log Returns
returns = np.log(df / df.shift(1)).dropna()

if len(returns) < 100:
    print("    Not enough data points for GARCH analysis.")
    return

# 2. Fit GARCH
print(f"    Fitting GARCH(1,1) to {asset1_name}...")
resid1 = fit_garch(returns[asset1_name])

print(f"    Fitting GARCH(1,1) to {asset2_name}...")
resid2 = fit_garch(returns[asset2_name])

# 3. Calculate DCC
print("    Calculating Dynamic Correlation...")
dcc = calculate_dcc_ewma(resid1, resid2)

# 4. Generate Figure
plt.figure(figsize=(12, 6))
plt.plot(dcc.index, dcc, label='DCC (GARCH-adjusted)', color='darkblue', linewidth=1.5)

rolling = returns[asset1_name].rolling(252).corr(returns[asset2_name])
plt.plot(rolling.index, rolling, label='Rolling Window (252d)', color='gray', linestyle='--', alpha=0.6)

plt.axhline(0, color='black', linewidth=1)

for name, (start, end) in CRISIS_PERIODS.items():
    plt.axvspan(pd.to_datetime(start), pd.to_datetime(end), color='red', alpha=0.1)

plt.title(f'Dynamic Conditional Correlation (DCC): {asset1_name} vs {asset2_name}',
fontsize=14)
plt.ylabel('Correlation')
plt.legend()
plt.tight_layout()

filename = f"figures/DCC_{asset1_name}_vs_{asset2_name}.png"
plt.savefig(filename, dpi=300)
print(f"    ✓ Saved Figure: {filename}")

# 5. Generate Summary Table
stats = {
    'Period': ['Full Sample'],
    'Mean Corr': [round(dcc.mean(), 3)],
    'Min Corr': [round(dcc.min(), 3)],
    'Max Corr': [round(dcc.max(), 3)],
    'Vol (Std Dev)': [round(dcc.std(), 3)] }

for name, (start, end) in CRISIS_PERIODS.items():
    mask = (dcc.index >= start) & (dcc.index <= end)
    sub = dcc[mask]
    if len(sub) > 0:
        stats['Period'].append(name)
        stats['Mean Corr'].append(round(sub.mean(), 3))
        stats['Min Corr'].append(round(sub.min(), 3))
        stats['Max Corr'].append(round(sub.max(), 3))
        stats['Vol (Std Dev)'].append(round(sub.std(), 3))

df_stats = pd.DataFrame(stats)
csv_name = f"tables/DCC_Stats_{asset1_name}_{asset2_name}.csv"
df_stats.to_csv(csv_name, index=False)
print(f"    ✓ Saved Table: {csv_name}")

if __name__ == "__main__":
    os.makedirs("figures", exist_ok=True)
    os.makedirs("tables", exist_ok=True)

    # Define pairs to analyze
    pairs = [
        ("Gold", "S&P 500"),
        ("Silver", "S&P 500"),
        ("Crude Oil", "S&P 500"),
        ("Copper", "S&P 500")
    ]

    for a1, a2 in pairs: generate_dcc_analysis(a1, a2)

```